

Planar Graph Metrics Embed into L_1 with Constant Distortion

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Abstract

We prove that every finite connected planar graph with arbitrary positive real edge lengths embeds into real L_1 with a universal distortion bound. This resolves the planar embedding conjecture positively.

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1 Introduction

Let $G = (V, E)$ be a finite connected undirected graph with positive real edge lengths. Its graph metric d_G is the shortest-path distance on V . An embedding of a finite metric space (V, d) into real L_1 has *distortion at most C* if, after multiplication by a positive scalar, its distances lie between d and Cd . The planar embedding conjecture asks whether one constant C works for every planar graph metric, independently of the number of vertices and the edge lengths. We prove that it does.

Theorem 1.1. *There is a finite universal constant $C \geq 1$ with the following property. For every finite connected planar graph $G = (V, E)$ with strictly positive real edge lengths, there exist a measure space $(\Omega, \mathcal{F}, \mu)$ and a map $f : V \rightarrow L_1(\Omega, \mathcal{F}, \mu; \mathbb{R})$ such that*

$$d_G(x, y) \leq \|f(x) - f(y)\|_1 \leq Cd_G(x, y) \quad (x, y \in V).$$

This proves the planar case of the conjecture of Gupta, Newman, Rabinovich, and Sinclair, which predicts uniformly bounded L_1 distortion in every proper minor-closed family [7]. In particular, the bound is independent of the number of distance scales in the graph metric.

1.1 Cuts, multicommodity flow, and the embedding problem

The relationship with cuts explains why L_1 is a natural target. For $S \subseteq V$, put $\delta_S(x, y) = |\mathbf{1}_S(x) - \mathbf{1}_S(y)|$. A nonnegative combination

$$D(x, y) = \sum_{S \subseteq V} w_S \delta_S(x, y), \quad w_S \geq 0, \tag{1}$$

is the ℓ_1 distance between the vectors $(w_S \mathbf{1}_S(x))_{S \subseteq V}$ and $(w_S \mathbf{1}_S(y))_{S \subseteq V}$. Conversely, threshold sweeps through real coordinates give such a representation for every L_1 metric on a finite set. Thus we seek weighted cuts that separate each pair in proportion to its graph distance while charging each edge at most a constant times its length. The edge bound then extends to every pair along a shortest path.

Linial, London, and Rabinovich established the connection between metric embeddings and multicommodity flow [14]; Aumann and Rabani independently obtained a general approximate

max-flow/min-cut theorem [2]. To state the corresponding planar consequence, give the edges nonnegative capacities and the unordered pairs of distinct vertices nonnegative demands, not all zero. Let λ_* be the largest common multiplier of the demands that can be routed fractionally. Let ϕ_* be the minimum ratio of crossing capacity to separated demand over cuts with positive separated demand. For $\lambda_* > 0$, the ratio ϕ_*/λ_* is the *flow-cut gap*. Gupta et al. showed that its worst value on a fixed graph equals the worst L_1 distortion over all weighted shortest-path metrics on that graph [7, Section 3, Theorem 3.2]. Hence Theorem 1.1 also gives a universal bound for the undirected fractional flow-cut gap on planar graphs, with arbitrary demands.

1.2 Earlier results and the remaining obstacle

The general preceding bound for n -vertex planar metrics is $O(\sqrt{\log n})$, due to Rao [18]. His construction embeds first into Euclidean space and then into L_1 ; it builds on the decomposition methods of Klein, Plotkin, and Rao for graphs excluding a fixed minor [9]. Newman and Rabinovich showed that even series-parallel metrics can require $\Omega(\sqrt{\log n})$ distortion into Euclidean space [15]. Thus a constant planar L_1 bound requires a different route. Abraham, Filtser, Gupta, and Neiman later recovered Rao’s planar bound through recursive shortest-path decompositions [1].

Constant distortion was known under several topological restrictions. Gupta et al. proved it for series-parallel graphs [7]. Chakrabarti, Jaffe, Lee, and Vincent obtained the sharp upper bound 2 [3], with a matching lower bound due to Lee and Raghavendra [12]. Chekuri, Gupta, Newman, Rabinovich, and Sinclair gave a bound exponential in the outerplanarity parameter [5]: the number of rounds needed to delete all vertices by successive outer-face removals in a planar drawing. This gives a uniform bound when that parameter is fixed.

Another line of work restricts the pairs whose distances must be preserved. The theorem of Okamura and Seymour, and its metric formulation by Hurkens, Schrijver, and Tardos, give an isometric L_1 representation of the metric on vertices of one face [16, 8]. If a terminal set is covered by γ faces in a fixed drawing, Krauthgamer, Lee, and Rika obtained a logarithmic bound in γ [10]. Filtser improved this to $O(\sqrt{\log(\gamma + 1)})$ [6]. Kumar constructed a single nonexpansive map on all vertices that preserves every cofacial pair within a universal factor, even when different pairs lie on different faces [11].

Geometric restrictions lead to a different family of results. Sidiropoulos proved constant distortion for planar metrics realized in simply connected surfaces of nonpositive curvature in the sense of Busemann [19]. Chalopin, Chepoi, and Naves proved that Busemann surfaces themselves embed isometrically into L_1 [4]. These are geodesic spaces homeomorphic to the plane whose distance is convex along pairs of geodesics.

The difficulty is to combine separation at different scales while keeping a uniform upper bound. Sidiropoulos addressed this difficulty by evolving a distribution of monotone cuts from coarse scales to fine scales [19, Sections 4–5]. Our construction also preserves monotonicity through successive local changes. For unrestricted planar metrics, two additional mechanisms provide the needed control: component separation limits which pairs need further updates, and interpolation makes local formulas agree exactly at their interfaces. The first yields a packing bound only for the surviving pairs; the second lets those local updates be combined without a loss proportional to the number of scales.

The structural reductions of Lee and Sidiropoulos relate the planar problem to the larger GNRS conjecture and a separate problem of closure under graph sums [13]. Together with the bounded-treewidth companion [17, Theorem 1.1], our theorem also gives constant distortion on their fixed-parameter almost-embeddable families; see Corollary 10.1. The general clique-sum

closure required for the full GNRS conjecture is not established here.

1.3 The construction and its main ideas

We construct two families of measurable label sets, denoted by $\mathcal{A}_v$ and $\mathcal{B}_v$ at a point v . A label determines the cut of points whose sets contain it, so $\mu(\mathcal{A}_x \triangle \mathcal{A}_y)$ is the resulting cut distance. The random construction chooses these entire families; the labels index their cuts. Every outcome has the required upper bound. Random local changes supply the lower bound.

Columns and the upper bound. Section 2 represents the geometric graph isometrically by vertical columns with zero-cost switches at specified heights. Choose a shortest-path tree rooted at a vertex and use distance from the root as height. Replace each non-tree edge by two ascending branches of the same total length and identify their tips. The plane embedding gives a circular order of columns in which the switching links do not cross. Figure 1 illustrates this realization.

For a fixed label, the active points on each column form an upper interval. Switches preserve membership, and the measure of labels whose membership changes along a vertical segment equals its length. Consequently every path bounds the cut distance between its endpoints, regardless of the number of updates. Section 3 formulates this invariant for both systems. The problem is to obtain lower separation while retaining it.

Common local formulas. Local updates use *charts*: regions on which one scalar field P and common label variables describe the thresholds. Their form is $h(v) + \sigma\kappa P(v)$, where h is height, σ is a fair sign, and $\kappa > 0$ is fixed. Averaging over the two signs detects both height differences and field differences. A sufficiently small vertical Lipschitz constant for κP preserves the upper-interval property.

Within a thin height band, use distances from a basepoint to the column midpoints to form successive radial layers. Noncrossing links restrict the entrances to a connected region beyond a layer boundary to neighborhoods of at most two points. This two-portal property, proved in Section 2, underlies the interpolation in Section 4. A field with Lipschitz constant strictly below one is approximated by pieces of the form $c \pm d_I(k, \cdot)$, where d_I is distance within the band and k is one point in it. At each entrance, the neighboring regions select exactly the same point-distance formula, including its source and its name. Strictness of the input slope bounds the number of candidate formulas that can govern a small ball. Fresh independent offsets then make a change of selected formula across that ball unlikely.

The first system selects the remaining pairs. At scale r , system A tries to raise the field near one endpoint of a pair at distance comparable to r , leaving the other endpoint unchanged. Section 6 shows that failure to produce a definite rise supplies a short path on which the old field decreases at almost the greatest rate its Lipschitz bound permits.

System A has a second separation mechanism. For each label, refine its active set into connected components of active columns. This refinement retains the upper bound. A sufficiently low inactive tip of a switching link separates active points on opposite sides and therefore certifies a lower bound. Only pairs with descending paths at both ends and without this certificate contribute *seeds*, the vertices near which the second system may update.

Proposition 6.6 places the seeds in a small narrow-band ball close to two short paths. Their distance from cell boundaries then separates their groups along those paths, bounding the number of groups that can affect one local update. Figure 3 shows the decisive case: when a path skips a

seed's column, failure of the low-tip certificate forces that seed close to the skipping link. The ambient metric need not have a uniform packing bound.

The second system separates the surviving pairs. The circular order supplies fields that increase along both arcs between two column indices. Section 7 constructs such a field below a point-distance function and equal to it at a chosen target. Equal values can occur on opposite arcs; a signed transverse field distinguishes the arcs and vanishes where a switch can join them.

Section 8 uses these fields in local changes near the seeds. Either a change can be admitted while agreeing with the old field at the boundary, or its failure in both orientations supplies short paths along which the ordered field rises and falls. If this happens at both endpoints, Lemma 7.6 puts them on opposite arcs, where a transverse increment separates them. The packing bound limits the number of competing randomized proposals, both for the separation probability and for chart persistence.

Sparse schedules preserve separation. Section 9 combines three estimates uniform in the graph and the update history. At scale r , a point's label set moves by at most Mr ; an eligible pair at distance in $[r, 2r)$ gains a separation certificate with fixed positive probability; and a ball of radius ρ already in one chart in each system is split between new charts with probability $O(\rho/r)$. Here eligibility requires neighborhoods of both endpoints, of radius a fixed multiple of r , to share the same chart data in each system.

Within a schedule, successive scales differ by a sufficiently large constant factor Q . The coarser chart-loss probabilities have a small geometric sum, so a pair reaches its own scale with positive probability. Later movement has another small geometric sum and cannot erase all the gained separation. For a component certificate, it suffices to preserve activity at the two endpoints and inactivity at one fixed tip. Averaging finitely many interleaved schedules covers every pair distance and completes the embedding.

2 Columns and thin bands

We first replace a plane graph by a family of vertical lines. This separates the length of a path from the planar restriction on its possible switches. The replacement preserves the original metric exactly. We then work in a thin height band and prove two estimates for the later local updates: entrances to a connected outer region lie near at most two fixed points, and within each radial layer, connected groups of columns admit short joining paths in the band. All these estimates use the width of the band and the circular order of the columns, with constants independent of their number.

Definition 2.1. A *column model* consists of a finite cyclically ordered set V_0 of indices, a copy $\{i\} \times \mathbb{R}$ of the height line for each $i \in V_0$, and a finite collection of links. A link ij has a prescribed closed bounded set $S_{ij} \subset \mathbb{R}$ of switch heights. At $H \in S_{ij}$ we identify (i, H) with (j, H) . The resulting quotient is denoted by X , and its height function by h . Paths consist of finitely many vertical segments and permitted switches; their length is the sum of the lengths of the vertical segments. The distance d is the infimum of these lengths.

The links are *noncrossing* if no two have four distinct endpoints alternating in the cyclic order. We require all column models below to have noncrossing links. We additionally require that every link between nonconsecutive indices is a switch at a single height H_e , and that neither endpoint

column has any switch above H_e . Such a switch will be called a *terminal switch*. Its quotient point is called the *tip* of the link.

For a closed interval $I \subset \mathbb{R}$, possibly unbounded, write $X_I = h^{-1}(I)$. The intrinsic distance d_I only permits paths with heights in I ; different components have distance $+\infty$. The active column graph G_I has vertex set V_0 and the links whose switch sets meet I as edges.

We write $B_{d_I}(x, r)$ for the closed ball and $d_I(x, S) = \inf_{z \in S} d_I(x, z)$ for distance to a set, with $d_I(x, \emptyset) = +\infty$. The same conventions apply to d .

Lemma 2.2 (Realization and shortest paths). *The function d_I is a metric on each component of X_I . Its finite distances are attained by paths using each column at most once. Components of X_I correspond exactly to components of G_I . If I is bounded, each component of X_I is compact. For arbitrary I , bounded closed subsets of each component are compact. In particular distance from a point to a nonempty compact set is attained.*

Proof. If a path leaves a column and later returns to it, replace the intervening part by the vertical segment joining its two visits. This does not increase length: total vertical variation is at least the difference of the two heights. The replacement stays in I . Repeating the operation leaves a path with no repeated column and at most $|V_0| - 1$ switches.

There are finitely many possible such column words. For a fixed word, its switch heights belong to a product of closed bounded sets $S_{ij} \cap I$. The length, including the initial and final vertical segments, is a continuous function of these heights. Thus every feasible word attains its minimum, and minimizing over the finitely many words proves attainment. A path of length zero has all its switches at the common endpoint height, so its endpoints are already identified in the quotient. This proves positive definiteness. The other metric axioms follow by reversing and concatenating paths.

A permitted path projects to a walk in G_I . Conversely any graph walk can be realized by choosing an available height for each edge and moving vertically between consecutive switches. This proves the component claim. The quotient map from any bounded closed piece of a column to X_I is 1-Lipschitz and hence continuous. Finitely many compact column pieces therefore have compact image. Finally $|h(x) - h(y)| \leq d_I(x, y)$, so a bounded metric set has bounded heights and is contained in such a compact image. A closed subset of that image is compact. $\square$

Proposition 2.3 (Planar graphs admit column models). *Every finite connected planar graph with positive edge lengths has an isometric realization of its geometric edge metric in a noncrossing column model satisfying the terminal-switch condition. The added parts of the model consist of separate infinite tails and do not shorten distances in the original graph.*

Proof. Fix a plane embedding and a root o . Let T be a shortest-path spanning tree and put $h(v) = d_G(o, v)$. Every tree edge is monotone in height and has length equal to the difference of its endpoint heights.

For each non-tree edge $e = uv$ of length ℓ_e , attach a new branch to u of length

$$\frac{\ell_e + h(v) - h(u)}{2}$$

and a separate new branch to v of length

$$\frac{\ell_e + h(u) - h(v)}{2}.$$

Both lengths are nonnegative because $|h(u) - h(v)| \leq \ell_e$. Their tips have the common height

$$H_e = \frac{\ell_e + h(u) + h(v)}{2}.$$

The enlarged abstract tree is denoted by $\tilde{T}$. Its branches may have zero metric length; retain their incidences and their plane order until taking the metric quotient. Identify the two tips belonging to each non-tree edge. This replaces that edge by two intervals of total length ℓ_e , with no additional identifications, and hence recovers the geometric graph metric.

Take one column for each root-to-leaf path of $\tilde{T}$. Place these columns in the cyclic contour order of its plane leaves. For neighboring leaves include a link on the heights of their common root prefix. These links make exactly the identifications prescribed by the tree. Indeed, at each tree point the descendant leaves form a consecutive block; adjacent leaves in that block identify all of its copies. Conversely neighboring leaves cease to be linked when their tree paths diverge.

For completeness, the relevant cyclic order can be obtained by taking a small closed disk neighborhood of a topological drawing of $\tilde{T}$. Draw each new branch as an initial portion of its corresponding non-tree edge. Zero metric lengths do not prevent drawing these portions as distinct topological arcs. The remaining portions of the non-tree edges are mutually disjoint arcs in the complementary disk on the sphere. Their endpoints on the boundary are the paired leaf tips. Two disjoint arcs in a disk cannot have alternating endpoints: one arc, together with either boundary arc joining its endpoints, separates the disk, and a second arc with alternating endpoints would have to cross it. Thus the tip links are noncrossing. The common-prefix links join cyclic neighbors and cannot cross any link.

Identify each pair of matching tips by its switch at H_e . A leaf column has no common-prefix switch above its own tip and belongs to no other tip pair. Consequently every non-neighbor tip link satisfies the stated terminal condition. Extend every column below the root and above its leaf by its own separate tail. These tails have no additional switches. An excursion into a tail must return through its point of attachment and can be deleted. It therefore cannot shorten any distance in the original geometric graph. The degenerate singleton case can be treated directly; a one-column tree is simply a path. $\square$

If the resulting model has one column, height embeds the original graph metric isometrically into $\mathbb{R}$; the singleton case is immediate as well. We henceforth assume that the column model has at least two columns.

2.1 Consequences of the circular order

The circular order gives separation statements used later for ordered fields and the seed packing argument. These statements concern the column graph, so they hold simultaneously for all height bands. Its outerplanarity will also give the two-portal estimate in the next subsection.

Lemma 2.4 (Chord separation). *Let uv be an edge of a noncrossing circular graph. Any graph path from the strict interior of one circular u - v arc to the strict interior of the other visits u or v .*

Proof. A path avoiding u, v and changing arcs has an edge whose endpoints lie in the two strict interiors. Its endpoints alternate with u, v , which contradicts noncrossing. $\square$

Lemma 2.5 (Gaps of a connected set). *Let S be a nonempty connected vertex set in a noncrossing circular graph. Every graph path avoiding S lies in one of the open circular gaps between consecutive vertices of S . Here the complement of a singleton is regarded as one gap. Connectivity of S and the path may be witnessed using different subgraphs of the same noncrossing graph.*

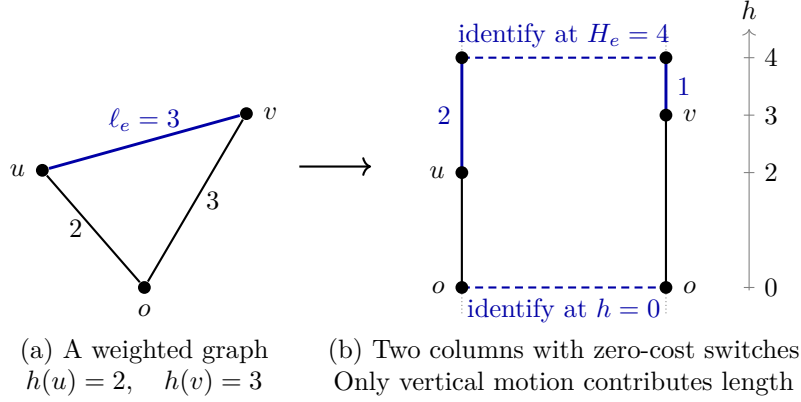


Figure 1: An exact example of Proposition 2.3. The black edges form a shortest-path tree rooted at o . The non-tree edge is replaced by two ascending twigs of lengths $(\ell_e + h(v) - h(u))/2 = 2$ and $(\ell_e + h(u) - h(v))/2 = 1$. Their tips have the same height $H_e = 4$ and are identified. Dashed horizontal segments denote identifications, not extra edges; the dotted vertical tails do not shorten original distances.

Proof. Suppose an edge uv of the path joined distinct gaps. Then each strict circular u – v arc contains a vertex of S . A path inside S joining these two vertices would have to visit u or v by Lemma 2.4. Both endpoints lie outside S , a contradiction. Thus each edge stays in one gap, and so does the whole path. $\square$

Lemma 2.6 (Skipping a column). *Give an open circular arc its linear order, and assume its complementary closed arc contains a column index. Let a vertex walk stay in the open arc, and let an index i lie strictly between its initial and final indices. Either the walk visits i , or one of its edges has endpoints strictly on opposite sides of i . In the latter case the edge is a non-neighbor link. If a second path starts strictly between the endpoints of this edge and leaves that strict interval, it visits an endpoint of the edge.*

Proof. If the walk avoids i , take its first edge changing from one side of i to the other. The index i lies on one strict circular arc between its endpoints, and an index in the complementary closed arc lies on the other. They are therefore not cyclic neighbors. The final assertion is Lemma 2.4. $\square$

Lemma 2.7 (Outerplanarity). *Every active graph G_I is outerplanar and has no $K_{2,3}$ minor.*

Proof. Place the indices on a circle and draw each edge as a straight chord. The noncrossing condition makes this an outerplanar drawing. Add a new vertex in the complementary disk on the sphere and join it to every column vertex by a disjoint radial star. The resulting cone graph is planar. If G_I had a $K_{2,3}$ minor, perform its deletions and contractions while retaining the new vertex and one edge from it to every contracted branch set. The resulting planar minor contains the cone over $K_{2,3}$. That cone contains $K_{3,3}$: add the cone vertex to the part of size two. This is impossible, since a simple bipartite planar graph on six vertices has at most eight edges, whereas $K_{3,3}$ has nine. $\square$

Lemma 2.8 (Connected distance sublevels). *Fix a component of X_I , a source o in it, and a height $H \in I$. Put $q_i = d_I(o, (i, H))$, taking $q_i = +\infty$ outside that component. For every real t , the nonempty set $\{i : q_i \leq t\}$ is connected in G_I .*

Proof. For an index i in the set, take a shortest path from o to (i, H) . If it visits a column j at height H' , stop there and move vertically to (j, H) . The cost of this new ending is $|H - H'|$, at most the vertical variation of the discarded suffix. Thus $q_j \leq q_i \leq t$. The projected path therefore lies in the sublevel set and joins i to the source column. Taking these paths for all i proves connectivity. This argument also shows that every source-column representative belongs to the sublevel whenever the sublevel is nonempty. $\square$

2.2 Portals and radial cells

We will cover the entrances to a connected radial superlevel set by at most two controlled balls. This estimate will then bound distances within the connected pieces of each radial layer.

In this subsection J is a closed band of width $a > 0$, H_* is its midheight, and we work in a fixed component of X_J . Choose a basepoint o and define

$$b_i = d_J(o, (i, H_*)).$$

For a point $v = (i, H)$ in the band,

$$|d_J(o, v) - b_i| \leq a/2, \quad |b_i - b_j| \leq a \quad \text{if } ij \in G_J. \quad (2)$$

The second estimate follows by traveling from the two midpoints to an available switch, at cost at most a . We occasionally use the weaker bound $2a$ without further comment.

Lemma 2.9 (Two portals). *Let C be a component of the subgraph of G_J induced by $\{i : b_i \geq t\}$. An entrance column of C is an index in C adjacent to an index with radial value below t . If entrance columns exist, there are at most two of their midpoints, called portals, such that every point on every entrance column lies within $200a$ in d_J of one of the portals. Every portal has radial value in $[t, t + a)$. If $t > a/2$ and C is nonempty, entrance columns exist.*

Proof. An entrance column has $t \leq b_i < t + a$ by (2). When $t < 30a$, all entrance midpoints have distance less than $31a$ from o . Any one is therefore within $62a$ of every other, and within $63a$ of every point on every entrance column. One portal suffices in this case.

Suppose $t \geq 30a$. Let L be the projection to columns of the closed ball $B_{d_J}(o, t - 12a)$. Shortest paths from o to the points of the ball stay in the ball, so L is connected. Every $j \in L$ has $b_j \leq t - 12a + a/2$, and hence L is disjoint from C .

Assume there are three entrance midpoints x_1, x_2, x_3 at pairwise distance greater than $100a$. Starting from each x_k , follow a shortest path toward o until its first visit to a column of L . This portion has length at most $13a$: by that length it reaches a point at radial distance $t - 12a$. Its projection joins C to L . The three projected portions are vertex-disjoint. Indeed a common column would permit a vertical transfer of length at most a , giving a path of length at most $13a + a + 13a < 100a$ between the corresponding entrance points.

For each portion retain the segment after its last visit to C and before its first visit to L . Its interior avoids both sets. It has an internal vertex, since its endpoints have radial values differing by at least $12a - a/2 > a$, while adjacent columns differ by at most a . Contract the connected sets C and L , and contract each of the three nonempty connector interiors to a separate vertex. This gives a $K_{2,3}$ minor, contrary to Lemma 2.7.

A maximal set of entrance midpoints at mutual distances greater than $100a$ thus has cardinality at most two. Maximality covers every entrance midpoint within $100a$, and the extra vertical distance is at most $a/2$. This proves the stated $200a$ bound. Finally a representative of o has radial value at most $a/2$. If $t > a/2$, any nonempty high component is a proper subset of the connected graph and has an entrance. $\square$

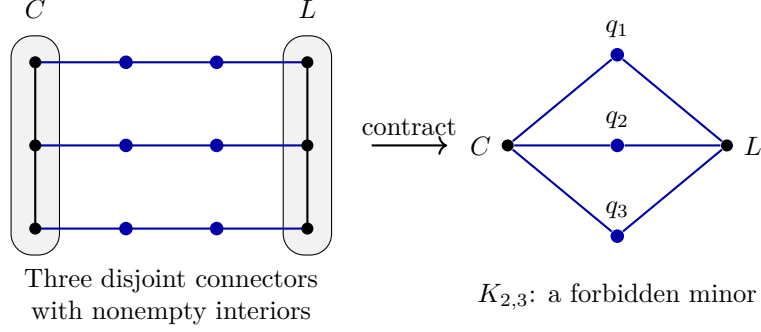


Figure 2: The hypothetical obstruction in Lemma 2.9. Three sufficiently separated entrance points would give vertex-disjoint paths from the connected high set C to the connected low set L . After trimming, each connector has a nonempty interior avoiding both sets. Contracting C , L , and the three interiors gives $K_{2,3}$, which cannot be a minor of the outerplanar column graph. The drawing records this contraction; it is not an outerplanar drawing of a configuration that the column model permits.

Definition 2.10 (Radial cells). Fix $D > 0$ and an offset $\theta \in [0, D)$. Partition the indices into layers by assigning i to the unique half-open interval

$$[\theta + nD, \theta + (n + 1)D), \quad n \in \mathbb{Z},$$

containing b_i . Split each layer into the connected components of its induced subgraph of G_J . A *cell* is the image in X_J of all intervals $\{i\} \times J$ belonging to one such component. Its *frontier* ∂C is the union of their images for indices adjacent in G_J to an index outside the cell.

This is a partition of the column indices, not necessarily a disjoint partition of their quotient images. If two cells share a quotient point, that point belongs to the frontier of each: a switch chain identifying representatives in the two cells has a first edge leaving either cell. Consequently formulas that agree with the unchanged data on all frontiers give unambiguous values at such shared points.

Here a *weak diameter* bound measures distances in the whole band; the paths witnessing those distances may leave the cell.

Lemma 2.11 (Diameter of cells). *Assume $a \leq D/1000$. Every radial cell is compact and path-connected, and has weak diameter at most $10D$ in d_J . Its frontier is compact, possibly empty.*

Proof. Each cell is the image of finitely many closed column intervals. Its column graph is connected, which makes that image path-connected. The same finite-union argument proves compactness of the cell and frontier.

Let q be the lower threshold of its layer. If $q \leq 30a$, every point of the cell has distance less than $D + 31a$ from o , so the diameter bound follows. Suppose $q > 30a$ and let C^+ be the component of $\{b_i \geq q\}$ containing the cell. It has at most two entrance portals by Lemma 2.9.

Take v in the cell and follow a shortest path to o until the first switch below threshold q . Before that switch the projected path stays in C^+ : two adjacent high columns belong to the same high component. The last high column is an entrance, and the switch point has radial distance at least $q - a/2$. Since $d_J(o, v) < q + D + a/2$, the portion from v to that point has length less than $D + a$. Some portal is within $200a$ of that point. Thus the cell is contained in the union of two closed balls of radius $D + 210a$ around its portals.

A connected set contained in the union of two closed balls either lies in one of them or meets their intersection. To see the second alternative when neither ball contains the set, intersect both balls with the connected set; two nonempty disjoint closed subsets could not cover it. In that case the centers have distance at most $2(D + 210a)$. In either alternative the cell has diameter at most $4(D + 210a) < 10D$. $\square$

Lemma 2.12 (Padding and separation of cell interiors). *Let $v = (i, H) \in X_J$, and let $T = \theta + D\mathbb{Z}$ be the radial threshold set. If*

$$\text{dist}(b_i, T) > \rho + 2a, \quad (3)$$

then every point reachable from v by a band path of length at most ρ has all its representatives in the same cell as i . Moreover $d_J(v, \partial C) > \rho$ for that cell.

For uniform $\theta \in [0, D)$, the probability that (3) fails is at most

$$\min\{1, 2(\rho + 2a)/D\}.$$

There is no factor for the number of representatives of v : their radial values differ pairwise by at most a .

In particular, if $a \leq D/1000$ and the radial values of two points are at distance greater than $0.02D$ from the thresholds, then their distances to their respective frontiers are greater than $0.01D$. If they belong to different cells, their mutual d_J -distance is greater than $0.01D$.

Proof. If a band path from v of length at most ρ visits column k , join its visit point to (k, H_*) and join v to (i, H_*) . The triangle inequality gives

$$|b_k - b_i| \leq \rho + a.$$

Thus every visited column stays in i 's layer under (3); its projected path puts it in the same connected component of that layer. The same argument applies after any zero-cost switch chain at an endpoint, proving the assertion about all representatives.

Every frontier column has its radial value within a of one of the cell's layer thresholds: its neighbor outside the cell cannot belong to the same layer. If z lies on a frontier column k , then

$$d_J(v, z) \geq |b_i - b_k| - a \geq \text{dist}(b_i, T) - 2a > \rho.$$

The formula remains true for an empty frontier with distance $+\infty$. The offset estimate is the length, at most $2(\rho + 2a)$, excluded in one period. Two representatives at height H have midpoint distance at most $2|H - H_*| \leq a$, since their points at height H are identical. This proves the representative assertion without a union bound.

For the numerical conclusion use $0.02D - 2a \geq 0.018D > 0.01D$. A path to a different cell must first visit a frontier column of the initial cell, so the same lower bound applies to its length. $\square$

The cell estimates will control the regions where the local updates act. For interpolation we also need the entrance estimate at successive radial thresholds. The resulting components form a rooted tree, with controlled distances between the portals at successive levels.

Lemma 2.13 (Successive portal levels). *Fix a step $\delta > 1000a$. At depth $n \geq 1$ take the components of $\{b_i \geq n\delta\}$ and give each its entrance portals from Lemma 2.9. The depth-zero node is the whole component, with its sole portal o . Each positive-depth node has a unique parent. For a node at depth n , every point on one of its columns with $b_i < (n+1)\delta$, and every portal of one of its children, is within $\delta + 220a$ of one of its own portals. Each child portal is at distance at least $\delta - 10a$ from every own portal.*

Proof. A connected component at a higher threshold is contained in a unique component at the preceding threshold, proving the parent assertion. For $n \geq 1$, repeat the root-geodesic argument in Lemma 2.11 at lower threshold $n\delta$. A point of the node's own layer has radial distance below $(n+1)\delta + a/2$. A child portal has radial distance in $[(n+1)\delta, (n+1)\delta + a)$. In either case its geodesic reaches an entrance of the parent node after length at most $\delta + 2a$, and an own portal is within a further $200a$. The asserted upper bound follows.

An own portal has radial distance in $[n\delta, n\delta + a)$, whereas a child portal has radial distance at least $(n+1)\delta$. Their distance is therefore at least $\delta - a$. At depth zero the upper bounds follow directly by taking a , and a depth-one portal is at distance at least δ from it. These estimates imply the stated bounds in every case. $\square$

3 Threshold sets and local coordinates

We construct cuts by assigning a measurable set of labels to every point of the column model. Along a column these sets will grow with height, with symmetric-difference measure equal to the vertical length. At a switch the sets will agree exactly. Every resulting cut distance is therefore bounded by the length of every column path. The local updates must produce lower separation while preserving these two properties.

There are two systems, called A and B . System A also uses the connected components of a label's active columns to separate points. System B supplies a second local separation test. Evolving distributions of monotone cuts through smaller scales is a central mechanism in Sidiropoulos's nonpositively curved planar construction [19, Sections 4–5]. Here the updates must also preserve the exact sets where local formulas meet.

3.1 Charts on a fixed label space

For each system fix a label space

$$(\Omega, \mu) = (\mathbb{R} \times \{-1, 1\}^{\mathbb{N}}, \mathcal{L} \otimes \nu),$$

where $\mathcal{L}$ is Lebesgue measure and ν is the product of fair sign measures. Every set in that system is a subset of this same Ω . The random choices of updates are independent of the label coordinates; we condition on those choices when describing the sets.

A *chart* consists of a region $\mathcal{R} \subset X$, a height band I containing its heights, a field P on the corresponding component of X_I (or on all of X_I), and measurable functions $Z : \Omega \rightarrow \mathbb{R}$ and $\sigma : \Omega \rightarrow \{-1, 1\}$ whose joint pushforward measure is Lebesgue measure times a fair sign. For $v \in \mathcal{R}$ set

$$\mathcal{A}_v = \{\omega \in \Omega : Z(\omega) < h(v) + \sigma(\omega)\kappa P(v)\}. \quad (4)$$

Write $\mathcal{B}_v$ for system B . The positive constant κ is fixed separately in each system; its values κ_A, κ_B will be chosen in Section 5.1.

At each stage there are finitely many chart identifiers. Each identifies common field and label data and their current region. These regions partition X , with a fixed rule for ties. A region may be disconnected; its connected pieces do not receive different identifiers. When some points receive a new chart, the old identifier is retained on the remaining region. Initially each system has one chart on X , with $P = 0$, Z the original Lebesgue coordinate, and σ the first sign coordinate.

Write $\text{Lip}_{\text{vert}}(P)$ for the supremum of the Lipschitz constants of P on its vertical column intervals. The following identity explains both the upper invariant and the purpose of changing the field.

Lemma 3.1 (Vertical control and rebasing). *For two points whose sets are expressed in common chart variables,*

$$\mu(\mathcal{A}_v \triangle \mathcal{A}_w) = \max\{|h(v) - h(w)|, \kappa|P(v) - P(w)|\}. \quad (5)$$

If $\kappa \text{Lip}_{\text{vert}}(P) < 1$ and P respects switches, the sets are nested increasingly on each column interval in the chart. Their symmetric-difference measure there equals the vertical length.

A branch $P(v) = \psi(h(v))$, where $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is Lipschitz with $\kappa \text{Lip}(\psi) < 1$, can be represented by a fresh constant chart without changing any sets.

Proof. For each sign, the symmetric difference of the two half-lines has measure $|\Delta h + \sigma \kappa \Delta P|$. Averaging the signs gives (5). The functions $H \mapsto H \pm \kappa P(i, H)$ are increasing, and their average increment is the height increment.

For rebasing put $\phi_\sigma(H) = H + \sigma \kappa \psi(H)$. These are increasing bijections of $\mathbb{R}$. Define the new coordinate on the same label space by $Z'(\omega) = \phi_{\sigma(\omega)}^{-1}(Z(\omega))$. Its pushforward measure is Lebesgue measure, since for $a < b$,

$$\mu\{a < Z' < b\} = \frac{\phi_+(b) - \phi_+(a) + \phi_-(b) - \phi_-(a)}{2} = b - a.$$

Choose a sign coordinate unused anywhere earlier in this execution. It is independent of Z' . Thus (Z', σ') has the required joint pushforward, and, for every label,

$$\{Z < \phi_\sigma(h)\} = \{Z' < h\}.$$

This describes exactly the same sets by a chart with field zero. $\square$

In an *ordinary chart*, the field has the form $P = c + \eta s$, where $\eta \in \{-1, 1\}$ and either $s(v) = d_I(o, v)$ for a fixed source o , or $s = 0$. For a constant field take $c = P$ and $\eta = 1$. Updating in *ordinary orientation* means changing s and converting back using the same c and η . System A always uses ordinary charts. System B begins with an ordinary chart and later uses the additional field type constructed in Section 7. Every recorded B -field will have Lipschitz constant less than 5 for finite d_I .

The interpolation in Section 4 will produce formulas for ordinary charts. These formulas carry names: when the same formula is inherited across an interpolation boundary, its name is preserved. When an interpolation formula becomes a chart, its new identifier records the old chart, the particular interpolation, and the selected formula's name, and retains the old label functions. Different old charts are not merged merely because their formulas agree. Within one old chart, recording an inherited formula therefore does not split a region merely because an internal interpolation boundary was crossed.

3.2 Exact gluing and component refinement

Every local update is first expressed in the old chart variables. A clipping replaces the old comparison field by its maximum with a proposal, which is the minimum of finitely many fields. Several proposals are combined by a finite maximum. On the boundary of each proposal's domain the proposal lies below the old field, with a strict margin in the spatial constructions below. Thus the clipping equals the old field at the endpoints of every updated column interval and at any switch leading outside the domain. Inside the domain all branches respect switches and have vertical Lipschitz constants whose products with κ are less than one.

The new sets consequently retain the global upper invariant. For points in one updated interval this follows from Lemma 3.1. For points in different updated intervals, compare the first

point with its interval endpoint, then use the old nested sets between the two interval endpoints, and finally compare with the second point. These three comparisons preserve nesting, and their symmetric-difference measures sum to the full vertical length. Unchanged points are handled by the same comparison. This argument does not require a finite number of updated intervals. Exact agreement at the boundary also preserves the sets at every switching identification. A Lipschitz perturbation that vanishes at the domain boundary admits the same interval-endpoint argument, provided the updated field respects switches and still has $\kappa \text{Lip}_{\text{vert}}(P) < 1$. The later increments in system B will satisfy these conditions.

Only after making the update do we record its winning formulas as new charts. A spatial branch inherits the old label functions Z, σ ; a height-only branch is rebased as in Lemma 3.1. An unchanged point keeps its old identifier. Height branches can have larger slopes than the recorded ordinary or B -fields; after rebasing their recorded fields are zero. Recording or rebasing these charts does not further change the newly assigned subsets of Ω . The identity (5) applies when the sets under comparison are written in common label variables; the global nesting and vertical-length invariant applies across all charts.

For a fixed label ω , call a point *active* when its assigned set contains ω . The active points of any column form an upper interval. Besides membership, we can use the connected components joined by active switches.

Lemma 3.2 (Component refinement). *Suppose a system of sets is nested increasingly on each column, respects switches, and has symmetric-difference measure at most vertical length. It defines an L_1 pseudometric dominated by d . It also admits the following refinement with the same upper bound.*

For a label ω , join two columns when one of their switches occurs at a height active for ω . To an active point assign the unit vector indexed by its component in this finite graph, and to an inactive point assign zero. Integrating the ℓ_1 distances of these vectors gives the refined pseudometric. It dominates the original symmetric-difference pseudometric.

If a non-neighbor link has inactive tip and two active points have representatives strictly on opposite sides of it, their refined distance for that label is 2.

Proof. For a fixed label all active points of a column have the same component coordinate. On a vertical segment the vector changes only when activity changes, and its ℓ_1 cost is the original indicator cost. At an active switch the two columns belong to the same component; at an inactive switch both vectors are zero. Integration and any finite path give the upper bound. A membership difference always costs one, proving domination.

Every switch incident to an endpoint column of a non-neighbor link is at or below its tip. If the tip is inactive, nesting makes all such switches inactive. A path of column indices between the two strict sides must pass through an endpoint of the link, by noncrossing. No active path can do so. The active points therefore have different unit coordinates.

There are only finitely many possible components, which can be indexed by subsets of the columns. An active link can be tested at its maximum switching height, so all coordinates are measurable. Subtracting the vector assigned to one fixed reference point makes each assignment integrable: the integrated norm of the difference is bounded by the distance to that point. The same subtraction works for the unrefined indicator system. $\square$

Only system A will be refined, once, after the last scale. We therefore track changes of membership during the construction, without tracking changes of component coordinates. The final component separation will be certified by activity at two vertices and inactivity at one fixed tip.

3.3 The estimates required of a local update

An update at scale $r > 0$ will try to separate original vertex pairs whose distances lie in $[r, 2r)$. At scale r , a pair (x, y) is *eligible* if the two closed balls $B_d(x, 5r)$ and $B_d(y, 5r)$ lie in one common old chart region in system A and in one common old chart region in system B . The common region may differ between the two systems. This condition supplies common field and label data for all short paths used in the pair's local tests.

The construction below will give constants $M, C_0 < \infty$ and $b, p, \rho_0 > 0$, independent of the graph and the update history, with three properties.

1. At every point of the full column space X , an update at scale r changes each unrefined label set by symmetric-difference measure at most Mr .
2. For a vertex pair with $d(x, y) \in [r, 2r)$, conditional on any past in which (x, y) is eligible, the update has probability at least p of producing either unrefined symmetric-difference separation at least br in one system, or a non-neighbor link separating representatives of x, y strictly and a system- A label set of measure at least br on which x, y are active and the link's tip is inactive.
3. Suppose a fixed ball $B_d(v, \rho r)$ lies in one old chart region in each system, where $0 < \rho \leq \rho_0$. Conditional on the past, the probability that this ball meets different new chart regions in either system is at most $C_0(\rho + \eta_0)$.

Here η_0 is a dimensionless upper bound for the relative mesh of the finite grids used to sample update parameters. It can be chosen arbitrarily small after the finite set of scales is fixed; the sampling convention is given in Section 5.2.

Propositions 9.1 and 9.2 establish these estimates after the local construction is complete. The third estimate keeps a pair eligible until its own scale with positive probability. The first then prevents sufficiently smaller scales from erasing the second estimate's gain. For a component certificate, this requires movement control at the tip as well as at the two vertices; that is why the first estimate concerns all of X . Section 9 combines the estimates by using schedules whose successive scales differ by a sufficiently large fixed factor.

4 Interpolation by single-source distance functions

All distances in this section are intrinsic to one connected component Y of a closed band X_J . We write d_J for this distance. The width of J is denoted by a , and its midpoint by H_* . The active column graph and its radial coordinate $b_i = d_J(o, (i, H_*))$ are as in Section 2; we choose o at height H_* . The choices of o and of all portals below are made before any random parameters in this section are sampled.

The construction will repeatedly use signed point-distance formulas as local data. We therefore approximate a function of Lipschitz constant strictly below one by formulas $c \pm d_J(k, \cdot)$, while controlling where the selected formula changes. Each formula has a fixed source and sign, a random additive constant, and a name. When the construction reuses a formula, it retains that name. Stability on a small ball means that one name is selected throughout the ball, not merely that the values agree at the interfaces.

Theorem 4.1 (Thin-band interpolation). *Let $0 < \zeta < 1$, let $E > 0$, and set*

$$\begin{aligned} \delta &= E/8, & a_0 &= 10^{-6}\zeta\delta, & q &= \zeta\delta/20, \\ \rho_0 &= \zeta\delta/1000, & M &= \lceil 15/\zeta \rceil + 3, & N &= 6(M + 1). \end{aligned} \tag{6}$$

Suppose $0 < a \leq a_0$, and let $p : Y \rightarrow \mathbb{R}$ be $(1 - \zeta)$ -Lipschitz for d_J . There is a random function $Q : Y \rightarrow \mathbb{R}$, obtained using finitely many independent scalar noises uniform on $[q, 2q]$, with the following properties.

1. For every realization, Q respects all switches, is 1-Lipschitz for d_J , and satisfies $|Q(v) - p(v)| \leq 6\delta < E$.
2. Its pieces use finitely many named atoms of the form

$$A(v) = C + \varepsilon d_J(k, v), \quad \varepsilon \in \{-1, 1\}, \quad k \in Y. \quad (7)$$

Each name specifies a source k and sign ε fixed before the noises are sampled; the intercept C is random. Whenever the construction reuses an atom, it retains its name.

3. For a fixed $x \in Y$ and $0 < \rho \leq \rho_0$, there is a list $\mathcal{A}(x, \rho)$ of at most N atom names and sources, determined without the noises, that contains the name of every atom selected by Q on $B_{d_J}(x, \rho)$. Except on an event of probability at most

$$\frac{N^2}{q}(8\rho + \eta), \quad (8)$$

one named atom is selected throughout this ball.

Here $\eta = 0$ for continuous uniform noises. Alternatively, each noise may be uniform on an equally spaced grid including both endpoints of $[q, 2q]$, and η in (8) may be the largest grid spacing. The same conclusions hold for any set of points reached from x by band paths of length at most ρ , including the points and representatives encountered on those paths.

4.1 The tree of superlevel components

For every integer $n \geq 0$, take the connected components of the subgraph on columns with $b_i \geq n\delta$. Its nonempty components are called nodes of depth n . The sole node of depth zero is the root. A node of depth $n + 1$ is contained in exactly one node of depth n , its parent. Because there are finitely many columns and all b_i are finite, there are only finitely many nodes. The layer belonging to a node ν of depth n consists of the full columns in that node satisfying

$$n\delta \leq b_i < (n + 1)\delta.$$

A node can have an empty layer; it is still retained in the tree. Every column has exactly one layer node.

The layers specify where a node's formula will be used. The formula itself will be defined on all of Y . Near an entrance to a child node, the child and parent formulas will agree exactly; this will allow us to paste their restrictions across layer boundaries without changing values or names.

For a positive-depth node, choose its set K_ν of one or two entrance portals using Lemma 2.9. The portals are midheight points on entrance columns. At the root set $K_\nu = \{o\}$. The radial and portal estimates of Lemma 2.13 give the following facts, which also apply to the root with this convention:

$$\begin{aligned} \min_{k \in K_\nu} d_J(k, v) &\leq \delta + 220a && \text{if } v \text{ is in the layer of } \nu \text{ or is a child portal,} && (9) \\ d_J(k, v) &\geq \delta - 10a && \text{if } k \in K_\nu \text{ and } v \text{ is a child portal.} && (10) \end{aligned}$$

For a positive-depth node of depth n , every $k \in K_\nu$ also satisfies

$$n\delta \leq d_J(o, k) \leq n\delta + 2a. \quad (11)$$

Every point on an entrance column is within $200a$ of K_ν . These are statements about d_J , so paths witnessing them may leave the corresponding node.

4.2 The formulas and their deterministic estimates

At each node we first approximate the input by positive distance cones based at its portals. We then constrain this tentative formula so that it agrees with the parent near every entrance. The strict Lipschitz bound on the input will limit how many generations an accurate atom can be inherited through.

For every portal $k \in K_\nu$, sample an independent noise $\alpha_{\nu k} \in [q, 2q]$. At every nonroot node also sample $\alpha'_{\nu k}, \alpha''_{\nu k} \in [q, 2q]$. All these noises are mutually independent. Define formulas on the whole of Y , recursively down the tree. The tentative formula is

$$T_\nu(v) = \min_{k \in K_\nu} (p(k) + d_J(k, v) + \alpha_{\nu k}).$$

At the root put $F_\nu = T_\nu$. This is 1-Lipschitz, because distance functions and their finite minima are 1-Lipschitz.

Now let ν have parent π , whose formula is already defined and 1-Lipschitz. To preserve the parent at the interfaces, we will constrain $T_\nu(v)$ to an interval containing $F_\pi(v)$ that collapses to this single value near K_ν . Define two envelopes of cones based at the parent values:

$$\begin{aligned} A_\nu(v) &= \min_{k \in K_\nu} (F_\pi(k) + d_J(k, v) - \alpha'_{\nu k}), \\ B_\nu(v) &= \max_{k \in K_\nu} (F_\pi(k) - d_J(k, v) + \alpha''_{\nu k}). \end{aligned}$$

If $d_J(k, v) < q/2$ for a portal k , the term with source k in the first envelope is below $F_\pi(v)$, and its term in the second is above $F_\pi(v)$. Indeed the parent changes by at most $d_J(k, v)$, whereas each noise is at least q . Thus $A_\nu(v) < F_\pi(v) < B_\nu(v)$ near every portal, even if the other portal's cones have different values. The desired interval is

$$[L_\nu(v), U_\nu(v)] = [\min(F_\pi(v), B_\nu(v)), \max(F_\pi(v), A_\nu(v))].$$

It always contains the parent value and reduces to that value in these portal neighborhoods. Clamp the tentative formula to this interval:

$$F_\nu(v) = \min(U_\nu(v), \max(L_\nu(v), T_\nu(v))).$$

All the formulas remain 1-Lipschitz on Y , since finite minima and maxima preserve this property. This completes the recursive construction.

The new atoms of a nonroot node are

$$p(k) + d_J(k, v) + \alpha_{\nu k}, \quad F_\pi(k) + d_J(k, v) - \alpha'_{\nu k}, \quad F_\pi(k) - d_J(k, v) + \alpha''_{\nu k}. \quad (12)$$

The root has its single tentative atom. An intercept $F_\pi(k)$ in (12) is a scalar, not an instruction to inherit the atom selected at k . Each expression in (12) is a new atom with source k . In contrast, selecting the branch $F_\pi(v)$ inherits its atom and its name.

Fix a total priority order on names in which all smaller-depth names precede larger-depth names. In every minimum or maximum, resolve equal values in favor of the earliest name. Multiple appearances of one inherited atom have the same name. This completely specifies a named atom selected by every formula at every point.

Lemma 4.2 (Accuracy and equality collars). *Every node formula F_ν has error at most 5δ at every portal of a child of ν , and error at most 6δ on its own layer. At a nonroot node with parent π , if*

$$\text{dist}_{d_J}(v, K_\nu) \leq c_0 := \zeta\delta/50,$$

then $F_\nu(v) = F_\pi(v)$, with the same selected atom name.

Proof. Since $c_0 < q/2$, the portal-neighborhood calculation above gives $U_\nu(v) = L_\nu(v) = F_\pi(v)$ in the stated collar. The priority convention retains the parent atom even if $T_\nu(v) = F_\pi(v)$.

For accuracy, the Lipschitz bound of p gives, at every point,

$$p(v) + \zeta \text{dist}_{d_J}(v, K_\nu) + q \leq T_\nu(v) \leq p(v) + (2 - \zeta) \text{dist}_{d_J}(v, K_\nu) + 2q. \quad (13)$$

In the situations of (9), its upper bound is less than $p(v) + 3\delta$, by (6). This proves both accuracy assertions at the root.

Proceed by induction. At a nonroot node, the inductive hypothesis gives $|F_\pi(k) - p(k)| \leq 5\delta$ for all its own portals. Therefore, writing $r_\nu(v) = \text{dist}_{d_J}(v, K_\nu)$,

$$A_\nu(v) \geq p(v) - 5\delta + \zeta r_\nu(v) - 2q, \quad (14)$$

$$B_\nu(v) \leq p(v) + 5\delta - \zeta r_\nu(v) + 2q. \quad (15)$$

At a child portal, (10) and $a \leq 10^{-6}\zeta\delta$ imply $\zeta r_\nu(v) \geq 2q$. Hence $U_\nu(v) \geq p(v) - 5\delta$ and $L_\nu(v) \leq p(v) + 5\delta$. Clamping a number in $[p(v), p(v) + 3\delta]$ to an interval with these two properties gives a value in $[p(v) - 5\delta, p(v) + 5\delta]$.

On the own layer, the same argument without the favorable distance term gives $U_\nu(v) \geq p(v) - 5\delta - 2q$ and $L_\nu(v) \leq p(v) + 5\delta + 2q$. Since $2q \leq \delta/10$, the clamped value has error at most $5.1\delta < 6\delta$. $\square$

Define Q on a representative (i, H) by the formula of the unique layer node of column i . Two columns joined by a switch have radial values differing by at most $2a < \delta$, so their layer depths are equal or consecutive. At equal depths their nodes coincide. At consecutive depths the upper node is a child of the lower node, and the upper column is an entrance column. Its switch point is within $200a < c_0$ of an upper portal, so Lemma 4.2 gives the same value and atom name from both representatives. Applying this to every switch also gives consistency under transitive identifications. Thus Q is defined on Y . It is 1-Lipschitz on each column and respects switches; summing along band paths shows that it is 1-Lipschitz for d_J . Its accuracy follows from Lemma 4.2.

4.3 One deterministic ancestor chain for a small ball

The pasted function is now well defined and has the required approximation error. To control changes of its selected formula, we next show that a small ball sees one node formula and only boundedly many of its ancestors. This step uses the geometry and the strict Lipschitz bound of the input; the random comparison estimate comes afterward.

Lemma 4.3. *For $x \in Y$ and $0 < \rho \leq \rho_0$, one node ν , determined without the interpolation noises, satisfies $Q = F_\nu$ on $B_{d_J}(x, \rho)$, including equality of selected atom names. If n is the depth of ν , every point of this ball has a representative on a column with $b_i \geq n\delta$.*

Proof. The closed ball is connected by geodesics through its center. Include all representatives of all its points in its column projection. For any two such representatives their radial values differ

by at most $2\rho + 2a < \delta$, using the radial comparison with $d_J(o, \cdot)$. Thus their layer depths take either a single value or two consecutive values $n, n+1$. The projected graph is connected, and all its columns have $b_i \geq n\delta$. They therefore belong to one superlevel node ν at depth n . In the single-depth case this proves the result.

Otherwise choose a represented point of layer depth n in the ball. For any represented point v of depth $n+1$, concatenate its geodesic to the center with a geodesic to the chosen point. This is a band path of length at most 2ρ . Its last crossing from depth n to depth $n+1$, when traversed toward v , enters the child node containing the representative of v . The entry column is an entrance column of that child. Hence v is within $2\rho + 200a < c_0$ of a portal of that child. The equality collar of Lemma 4.2 identifies its formula and its selected atom with those of ν at v . This proves the assertion for every representative. All the choices in this argument depend only on the band geometry and the ball. $\square$

Lemma 4.4 (Bounded depth of a selected atom). *For the node ν supplied by Lemma 4.3, every atom selected by F_ν on the ball originates in ν or one of its last M ancestors. Consequently at most N atom names can be selected there.*

Proof. We first prove an obstruction to passing a parent atom through a distant node μ . Suppose

$$\zeta d_J(k, v) > 13\delta \quad \text{for every } k \in K_\mu. \quad (16)$$

Equations (13)–(15) give

$$T_\mu(v) > p(v) + 13\delta, \quad A_\mu(v) > p(v) + 7\delta, \quad B_\mu(v) < p(v) - 7\delta.$$

If the parent value belonged to $[p(v) - 6\delta, p(v) + 6\delta]$, then $L_\mu(v) = B_\mu(v)$ and $U_\mu(v) = A_\mu(v)$, so

$$F_\mu(v) = \min(A_\mu(v), T_\mu(v)) > p(v) + 7\delta.$$

In particular, a value within 6δ of $p(v)$ cannot be obtained at this node by selecting its parent atom. Notice that the lower bound on the tentative formula is required here as well as the two clamping bounds.

Let n be the depth of ν . If $n \leq M$, every ancestor is already in the asserted list. Otherwise consider its ancestor μ of depth $n - M \geq 1$. For a point v of the ball, Lemma 4.3, radial comparison, and (11) give

$$d_J(k, v) \geq M\delta - 3a \quad (k \in K_\mu).$$

The definition of M implies (16). An atom originating strictly earlier than μ could be selected by $F_\nu(v)$ only by being inherited through every intervening node. All these inherited values would equal $F_\nu(v) = Q(v)$, whose error is at most 6δ . In particular μ would select its parent atom with such a value, contradicting the preceding paragraph. This excludes all earlier origins. Each of the remaining $M+1$ nodes contributes at most six atoms, proving the count N . $\square$

4.4 Conditional comparisons and probability bounds

We have a list of at most N possible names on the ball, fixed before the noises are sampled. Their intercepts can depend on earlier noises. The next lemma identifies an independent noise in each comparison, so this dependence does not require a union bound over the whole node tree.

Lemma 4.5 (Fresh-noise comparison). *For any two distinct named atoms in the deterministic list of Lemma 4.4, any fixed $x \in Y$, and $t \geq 0$,*

$$\Pr(|A(x) - A'(x)| \leq t) \leq (2t + \eta)/q. \quad (17)$$

The same bound holds when one atom is replaced by any real quantity independent of all current interpolation noises.

Proof. For two atoms originating at different depths, use the later atom's own noise. Its intercept, whether $p(k)$ or $F_\pi(k)$, does not depend on that noise, because parent formulas use only earlier nodes. The earlier atom does not depend on it either. Thus, after conditioning on every other noise, the difference has the form $c + \varepsilon\alpha$ with $\varepsilon \in \{-1, 1\}$ and α uniform on $[q, 2q]$. For two atoms at the same node their own noises are different and their parent intercepts depend on neither, so the same argument applies. The candidate list lies on one ancestor chain, so these are all cases. An independent comparison quantity can likewise be included in the conditioning.

For a continuous uniform noise an interval of length $2t$ has probability at most $2t/q$. For a grid with spacing $\eta' \leq \eta$, it contains at most $2t/\eta' + 1$ grid points; division by the total number $q/\eta' + 1$ gives at most $(2t + \eta')/q$. $\square$

Completion of Theorem 4.1. Take the deterministic list furnished by Lemma 4.4. The union bound and Lemma 4.5, with $t = 4\rho$, show that the probability of a pair satisfying $|A(x) - A'(x)| \leq 4\rho$ is at most $N^2(8\rho + \eta)/q$. On the complementary event, two atoms in the list cannot be equal anywhere on the ball, since both are 1-Lipschitz. The continuous function $Q = F_\nu$ selects a value from this finite list at every point. Where the list values are distinct its selected name is locally constant: continuity and the positive minimum gap to the other finitely many values give a neighborhood with the same selection. A ball is connected by paths through its center, so the name is constant throughout the ball. This proves (8). The path version follows by applying the ball assertion to the paths, all of whose points lie in the ball. $\square$

Corollary 4.6 (Comparison with other branches). *Condition on arbitrary prior data. Let W be the union of a nonempty family of band paths issuing from a fixed point x , each of length at most ρ . The set W and these paths are fixed by the prior data. Suppose that $Q_1, \dots, Q_s$ are interpolation constructions, each on a band component containing all these paths. Their geometries, inputs, and portals are fixed by the prior data, and their fresh noise families are mutually independent. The bands and their intrinsic metrics may differ; each of the specified paths must be permitted in every band. Denote the constants of Q_i in (6) by $q_i, N_i, \rho_{0,i}$.*

Allow inputs $c_i + \varepsilon_i Q_i$, with $\varepsilon_i \in \{-1, 1\}$ and c_i fixed by the prior data. Also allow one fixed branch f and t branches $g_j + \beta_j$, where f and the g_j are fixed functions whose restrictions to each specified path, parametrized by arclength, are L -Lipschitz. The β_j are fresh independent uniform noises on intervals of length $w_j > 0$, independent also of the interpolation noises. All functions respect the switches used by the paths. Set

$$L_* = \max(1, L), \quad S = 1 + t + \sum_{i=1}^s N_i, \quad q_* = \min(\{q_i : 1 \leq i \leq s\} \cup \{w_j : 1 \leq j \leq t\}).$$

Assume that there is at least one randomized input and that $\rho \leq \min_i \rho_{0,i}$ when $s > 0$. Any fixed expression formed from these inputs by finitely many minima and maxima selects one branch and, for a selected interpolation, one atom throughout W , except with conditional probability at most

$$\frac{S^2}{q_*}(8L_*\rho + \eta). \quad (18)$$

Continuous noises have $\eta = 0$; for equally spaced endpoint grids η can be the largest spacing of any noise used here. The expression uses each interpolation with a single chosen sign and shift; repeated appearances of that same input are given the same name.

Proof. Every specified path lies in $B_{d_{J_i}}(x, \rho)$ for each interpolation band J_i . Use the deterministic bounded atom list for Q_i on that ball, with its chosen sign and shift. Adjoin the shifted g_j branches and, if present, f . For pairs from one interpolation use Lemma 4.5. For pairs from different interpolations condition on all noises except the own noise of an atom in one interpolation; independence of the constructions makes the other atom independent of that noise. The same argument applies to a shifted g_j or the fixed branch f . There is at most one branch without a fresh noise. Therefore every comparison of distinct names at x has the interval bound $(2u + \eta)/q_*$ at tolerance u .

Take $u = 4L_*\rho$ and sum over at most S^2 pairs. Off the event in (18), no two listed values agree anywhere on W : their changes along each path from x are bounded by $L_*\rho$. The expression is a continuous selection from the resulting finite list along each path. Its selected name is therefore constant on each path and, because all paths begin at x , throughout W . $\square$

Remark 4.7. The conditional independence requirements in Corollary 4.6 are substantive. In particular, a branch defined using current interpolation noises cannot be treated as a fixed comparison branch, and two arbitrary fixed branches need not have a small probability of meeting. The later applications use one old branch, plus independently shifted height or ordered branches, after conditioning on the bands, old charts, groups, and admissions. All choices of geometric portals and interpolation inputs precede the noises being tested. With this conditioning, the displayed bounds are uniform in all prior choices and in the size of the column graph.

5 Scales and the local experiment

We now specify the scales, geometric randomizations, and common inputs for the local updates. The next sections construct those updates and prove the separation and persistence estimates stated above.

5.1 Fixed constants and scales

The geometric constants satisfy $\epsilon \ll j \ll g \ll R \ll D \ll 1$. The table records their roles; the explicit choices below supply every numerical inequality used later.

Parameters	Role
D, R, g	Main radial cell width, secondary update radius, and radial width for grouping seeds.
j, ϵ	Detectable drop at a separating tip, and error in a descending path.
a_1, a_2	Broad and narrow band widths; control vertical transfers and interpolation.
E_A, E_B	Interpolation errors in the two systems.
κ_A, κ_B	Convert field variation into label mass while preserving vertical nesting.
λ	Size of the auxiliary correction in system B (Section 8.2).
Q	Scale ratio; controls earlier chart splitting and later movement.

Fix the following numerical constants:

$$R = 10^{-8}, \quad D = 10^{-3}, \quad g = 10^{-14}, \quad j = 10^{-20}, \quad \epsilon = 10^{-24}, \quad E_A = \epsilon/1000, \quad E_B = 10^{-3}R.$$

Set $\zeta_A = \epsilon/100$ and $\zeta_B = 1/10$. Choose $a_1 > 0$ within the width bound of Theorem 4.1 for (ζ_A, E_A) , within the bounds of Lemmas 2.11 and 2.12 for D and g , and with $a_1 \leq 10^{-6}E_A$. Set $\kappa_A = a_1/10^6$. Choose $a_2 > 0$ within the bound of Theorem 4.1 for (ζ_B, E_B) and with

$$a_2 \leq 10^{-6} \min\{a_1, \kappa_A j\}, \quad \kappa_B = a_2/10^6.$$

Reserve $K = 100$ as a Lipschitz bound for the auxiliary field in system B ; Lemma 8.3 will supply this bound. Set $\lambda = R/(10^{12}K)$. If necessary decrease a_1, a_2 further to meet the explicit thin-band hypotheses of the geometric lemmas, and then choose their dependent constants in the order just specified. All inequalities in this choice are upper bounds on a newly chosen positive number. The scale spacing $Q = 2^m$ is chosen last.

Choose finitely many dyadic scales so that every nonzero vertex distance belongs to $[r, 2r)$. For each residue class of exponents modulo m , run one schedule in decreasing order of its scales. For the analysis of a step, divide distances, heights, fields, and their additive constants by its scale r . Multiply the resulting threshold and field changes by r before evaluating them in the original label space. In particular a field change ΔP in scale units moves label mass $r\kappa|\Delta P|$. The label coordinates are not resampled or renormalized.

5.2 The data of a step

At a step tile the height line by intervals J_1 of width a_1 with a common random shift, and independently by intervals J_2 of width a_2 . For the two clipping constructions, process every broad or narrow band meeting the height interval

$$[\min_{v \in V} h(v) - 6, \max_{v \in V} h(v) + 6]$$

in the current scale units, and make no clipping update in other bands. This is a finite collection and contains every band needed for a radius-5 neighborhood of an original vertex. The decision whether to process a band is common to that entire band. This convention does not truncate the ordered increments of Section 8.2, which act on their current chart regions. Given the bands, fix the radial basepoints in their components before sampling any radial layer origins. In every broad-band component form both the width- D radial cells and the width- g radial cells, using independently shifted origins. The narrow bands will instead determine the domains of the secondary proposals and the balls in the packing estimate. Interpolation basepoints and portals are likewise fixed before any interpolation noises.

A vertex is *interior* if its narrow band is contained in its broad band, its height avoids the outer tenths of both bands, and all its column representatives have radial values farther than $0.02D$ and $0.02g$ from the respective layer interfaces. A pair with height difference less than $10^{-4}a_2$ has probability at least $1/4$ of satisfying these requirements in common bands. Indeed each of the two height tests has failure probability at most 0.21 , and each radial test, for two vertices together, has failure probability at most 0.081 , with the remaining allowance covering the column-width and grid errors. A chosen representative controls all the others within an error at most a_1 , without a union bound over columns. Interior in the broad band implies containment of the narrow band because $a_2 \leq 10^{-6}a_1$.

Recall that eligibility at the current scale requires the two closed d -balls of radius 5 to lie in one common old region in each system. At this scale use the complete finite list of eligible original vertex pairs with $1 \leq d(x, y) < 2$. For the detailed tests impose

$$|h(x) - h(y)| < 10^{-4}a_2, \quad |s_A(x) - s_A(y)| < E_A, \quad |P_B(x) - P_B(y)| < 10^{-4}R. \quad (19)$$

Here s_A is the ordinary orientation and P_B is the full old field. Failure of a cutoff already gives positive separation by (5), so it is handled by the no-change mode.

There are four equally likely modes: no change, the main update in system A , the secondary clipping in system B , and the ordered increment in system B . The surviving groups and admission tests will be defined in Sections 6.2 and 8. These lists, tests, and geometric data use the old charts; they are determined independently of the choice of mode and the fresh interpolation noises. Cell and group coins are independent fair coins. Where the secondary construction in Section 8 calls for a fair sign, all points with one old chart identifier use the same sign. These construction signs are independent of the label-coordinate sign σ .

Thus a step first determines its bands and cells, then its eligible pairs, surviving groups and admissions, and finally its fresh interpolation and branch offsets. Conditioning on the earlier choices leaves those last offsets independent. In particular, no admission decision is made by examining a newly selected interpolation atom.

To avoid selection-measurability issues, all bounded-interval geometric randomizations may use finite equally spaced grids. For a fixed finite set of scales take their relative meshes at most η_0 , where η_0 is smaller than every finer-to-coarser scale ratio and small enough for the fixed interior-probability allowances above. An interval probability bound $C\rho$ becomes $C'(\rho + \eta_0)$, which has the required size for all comparisons in the proof. There are finitely many outcomes at every step and finitely many nodes in each interpolation hierarchy. Choices depending on past outcomes can therefore be made separately on each branch of a finite tree. This makes every final random mixture a finite direct sum.

6 The main update and a packing bound for secondary seeds

This section gives system A its two separation mechanisms and extracts the geometric information available when both fail. Main clipping either separates a pair or supplies short descending paths at its endpoints. A low separating tip supplies a component certificate. The remaining endpoints satisfy a packing bound that lets one secondary proposal act alone with positive probability and bounds the number of formulas competing near a point.

All lengths in this section are measured in the units of the current scale. Thus the pairs under consideration satisfy $1 \leq d(x, y) \leq 2$. We use the parameters and the random bands of Sections 5.1 and 5.2. In particular, $J_2 \subset J_1$ denotes a narrow band contained in its broad band. The cells used below are the connected column cells of Lemma 2.11. The frontier of a cell C is the union, within J_1 , of all its column intervals whose columns have an active link to a column outside C . When two cells have a common quotient point, that point is in the frontier of both cells. Frontier values will therefore always be left unchanged.

An ordinary chart of system A has field $P = C_0 + \tau s$, where $\tau \in \{-1, 1\}$ and $s = d_I(o, \cdot)$ on a component of an old band I , or $s = 0$ in the constant case. Within that chart we perform all comparisons in the s orientation. Replacing s by s' means replacing P by $C_0 + \tau s'$ in the original label variables.

6.1 The main update

Let C be a cell of width D in J_1 . We make it available for an update only when its entire closed d -neighborhood of radius 1 is contained in one old ordinary chart of system A . This condition allows every path of length at most $10D$ starting in C to be evaluated in that chart: all its points are in the indicated neighborhood, and hence in the band component on which the old field is defined. In particular,

$$|s(v) - s(w)| \leq d_{J_1}(v, w) \leq 10D \quad (v, w \in C). \quad (20)$$

Here and below an intrinsic shortest path of length at most $10D$ is allowed to leave the cell. Its length bound, rather than intrinsic connectedness of the cell, ensures its availability in the old chart.

The frontier bounds how far a strictly Lipschitz proposal can rise at an interior point. If a frontier constraint prevents a definite rise, a shortest path to that constraint will be almost maximally descending for the old field. A downward shift makes the proposal strictly inactive at the frontier, which will also protect small balls crossing cell boundaries.

Fix $v_C \in C$ and put $s_C = s(v_C)$. On the entire J_1 -component containing C , define

$$p_C(v) = \min \left\{ s_C + 1, \inf_{z \in \partial C} (s(z) + (1 - \zeta_A)d_{J_1}(v, z)) \right\}. \quad (21)$$

The infimum over an empty frontier is $+\infty$. The frontier is a finite union of compact column intervals, so that a finite infimum is attained. The function p_C is $(1 - \zeta_A)$ -Lipschitz for d_{J_1} . Apply Theorem 4.1, with error E_A , to obtain $\tilde{p}_C$, and set

$$F_C = \tilde{p}_C - 4E_A. \quad (22)$$

In particular,

$$p_C - 5E_A \leq F_C \leq p_C - 3E_A. \quad (23)$$

For a fresh offset $\alpha_C \in [0, E_A]$, define a function of height $p_{h,C}$ as follows. It is $s_C - 1 + \alpha_C$ at either endpoint of J_1 , is $s_C + 2 + \alpha_C$ on the interval obtained by deleting the outer portions of width $0.09a_1$, and is linear on the two remaining portions. Extend it constantly beyond the two endpoints of J_1 . Its Lipschitz constant is $3/(0.09a_1)$. If the independent fair coin of C is kept, make the replacement

$$s'(v) = \max\{s(v), \min\{F_C(v), p_{h,C}(h(v))\}\} \quad (v \in C). \quad (24)$$

Otherwise leave the cell unchanged. Cells that are unavailable are also left unchanged.

Lemma 6.1 (Validity and size of the main replacement). *The simultaneous replacements (24) respect switches, preserve vertical nesting and the vertical upper bound for system A , and admit ordinary charts on every new branch. At any point the change in the system- A label set has measure at most $\kappa_A(1 + 10D)$.*

Proof. At a frontier point z , the term with source z in (21) gives $p_C(z) \leq s(z)$. Thus $F_C(z) \leq s(z) - 3E_A$. At either endpoint of the height band, (20) gives

$$p_{h,C} \leq s - 1 + 10D + E_A < s.$$

The replacement therefore equals the old field on every frontier and at every height endpoint, with a strict margin in the proposed branch. Different cell prescriptions agree at common quotient points. Within a cell, every constituent respects switches.

The old field and F_C have vertical Lipschitz constant at most 1; the height branch has constant $3/(0.09a_1)$. The maximum and minimum in (24) have the maximum of these constants. The parameter choice makes its product with κ_A strictly less than 1. The vertical replacement and gluing rule for label sets consequently preserves nesting and the vertical upper bound.

An unchanged branch keeps its old ordinary chart. A winning atom of F_C has the form $c \pm d_{J_1}(k, \cdot)$ and therefore gives an ordinary chart after converting back from the s orientation. A winning height branch is recorded by the height-only change of variables from the label-set construction, with a fresh constant chart. Identifiers can be chosen using the update data and the winning atom, with the inherited identifier retained wherever the old field wins.

Finally, the replacement never decreases s , and

$$F_C \leq s_C + 1 - 3E_A \leq s + 1 + 10D.$$

Consequently $0 \leq s' - s \leq 1 + 10D$. At a fixed point, a change $s' - s$ in the common chart changes the label set by measure $\kappa_A |s' - s|$, which proves the stated bound. $\square$

Definition 6.2 (Descending path). A descending path at a point w is a shortest path in J_1 , of length $d_0 \in [10^{-5}, 0.02]$, whose terminal point z satisfies

$$s(z) \leq s(w) - d_0 + \epsilon. \quad (25)$$

It is parametrized by arclength $t \in [0, d_0]$. All descending paths used below lie in the old ordinary chart band, so s can be evaluated at every point of the path.

Lemma 6.3 (Main raise or a descending path). *Suppose w satisfies the height and radial interior requirements for its width- D cell, and that the cell is available. For every choice of the interpolation parameters, its kept replacement raises $s(w)$ by more than $2E_A$, unless a descending path at w exists. Every descending path satisfies*

$$s(\gamma(t)) \leq s(w) - t + \epsilon \quad (0 \leq t \leq d_0). \quad (26)$$

Proof. Assume that the raise is at most $2E_A$. The height interior requirement places w on the plateau of $p_{h,C}$, where

$$p_{h,C}(h(w)) \geq s(w) + 2 - 10D > s(w) + 2E_A.$$

It follows that $F_C(w) \leq s(w) + 2E_A$, and (23) then gives

$$p_C(w) \leq s(w) + 7E_A. \quad (27)$$

The constant branch $s_C + 1$ is strictly greater than the right-hand side of (27). The frontier is therefore nonempty, and there is a minimizing $z \in \partial C$ with

$$s(z) + (1 - \zeta_A)d_{J_1}(w, z) \leq s(w) + 7E_A.$$

Put $d_0 = d_{J_1}(w, z)$. The radial padding conclusion of Lemma 2.12 gives $d_0 \geq 0.01D = 10^{-5}$. The weak diameter estimate gives $d_0 \leq 10D < 0.02$. Choose an intrinsic shortest path from w to z . It remains in the old chart band by availability. Since

$$\zeta_A d_0 + 7E_A \leq (\epsilon/100)(10D) + 7\epsilon/1000 < \epsilon,$$

this path satisfies (25).

For the last assertion, apply the old field's Lipschitz bound to the remaining subpath, of length $d_0 - t$:

$$s(\gamma(t)) \leq s(z) + (d_0 - t) \leq s(w) - t + \epsilon.$$

□

Corollary 6.4 (Separation supplied by the main test). *Let (x, y) be an eligible pair satisfying the interior requirements, $1 \leq d(x, y) \leq 2$, and $|s(x) - s(y)| \leq E_A$. If either endpoint has no descending path, then, conditional on main-update mode, the independent cell coins give probability at least $1/4$ of system- A symmetric-difference separation at least $\kappa_A E_A$ at this step.*

Proof. The two endpoints lie in different width- D cells, since a cell has d -diameter at most $10D < 1$. Each of these cells is available: its radius-1 neighborhood lies in the radius-5 neighborhood of its endpoint, by the cell diameter bound and eligibility. Keep the coin at an endpoint with no descending path and suppress the coin of the other endpoint. This has probability $1/4$. Lemma 6.3 raises the first comparison value by more than $2E_A$ and leaves the second unchanged, so their new field difference has magnitude at least E_A . Evaluate both new sets in their common old label variables. The common-chart symmetric-difference formula gives the asserted bound. □

6.2 The component barrier and the seed set

The main test has reduced the problem to pairs with two descending paths. Before using those paths to locate nearby groups, we remove pairs already separated by a low terminal tip. The resulting certificate records only three memberships, so later changes elsewhere cannot invalidate it.

For the rest of the section, retain only eligible pairs (x, y) in the distance range $[1, 2]$ that satisfy all the interior requirements and both field-difference cutoffs, lie in the same bands $J_2 \subset J_1$, and possess descending paths at both endpoints. In particular, their common ordinary system- A field s is nonconstant: a constant field cannot satisfy (25) with $d_0 \geq 10^{-5} > \epsilon$. Choose once and for all a representative column for every vertex under consideration. A chord separates two representatives strictly if they lie in the strict interiors of the two complementary circular arcs determined by its endpoint columns.

Lemma 6.5 (Label mass at a separating tip). *Suppose a retained pair has a non-neighbor link separating its representatives strictly, with tip e at a height in J_2 , such that $d(e, x) \leq 1$ or $d(e, y) \leq 1$. If*

$$s(e) < s(w) - j \tag{28}$$

for at least one endpoint w , then there is label mass at least $\kappa_A j/4$ for which both endpoints are active and e is inactive. In no-change mode this gives a component-refined separation in system A of at least $\kappa_A j/2$ at this step.

More generally, suppose subsequent changes have total unrefined symmetric-difference costs $\Delta_x, \Delta_y, \Delta_e$ at the three points. The final component-refined separation is at least

$$2(\kappa_A j/4 - \Delta_x - \Delta_y - \Delta_e)_+. \tag{29}$$

Proof. Eligibility puts x, y, e in the same old ordinary chart. Write $P = C_0 + \tau s$ there and restrict labels to the sign $\sigma = \tau$. On this half of the label space the threshold at v is

$$T_v = h(v) + \kappa_A \tau C_0 + \kappa_A s(v).$$

The other endpoint has s -value within E_A of $s(w)$. Also all three heights lie in J_2 . Thus, for $v = x, y$,

$$T_v - T_e \geq \kappa_A(j - E_A) - a_2 \geq \kappa_A j/2,$$

using $E_A \leq j/4$ and $a_2 \leq \kappa_A j/4$. The labels with $\sigma = \tau$ and $T_e \leq Z < \min\{T_x, T_y\}$ have the required three membership conditions and mass at least $\kappa_A j/4$. Choices at the threshold endpoints have measure zero.

Lemma 3.2 separates the two active points into different components whenever the tip is inactive; each such label contributes 2 to the refined distance. After later changes, the subset of this label mass on which any of the three memberships changes has measure at most $\Delta_x + \Delta_y + \Delta_e$. Apply the same component-refinement lemma to the surviving labels in the final nested system. This proves (29); no control of the changes in component names is required. $\square$

Exclude every retained pair to which Lemma 6.5 applies. The secondary seed set consists of all endpoints of the remaining pairs. If a vertex occurs in several remaining pairs, regard it as one seed and fix one of those pairs as its witness; write w' for its witness partner. Fix descending paths for both members of each chosen witness pair. All these choices are made from the old fields and bands, before any secondary interpolation noises are sampled.

Group seeds by their broad and narrow bands, their width- g cell in the broad band, and their old chart identifiers in both systems. The groups are used only to organize the update. They introduce no new chart boundaries. By the weak cell diameter bound, two seeds in the same group have d_{J_1} -distance at most $10g$; a shortest path of this length is in their common old chart bands by eligibility. Their ordinary fields consequently vary by at most $10g$. The full system- B fields vary by at most $50g$, since their old-chart Lipschitz constants are less than 5.

6.3 Packing the groups that meet a narrow-band ball

We put the relevant seeds in one circular gap of a connected low sublevel set. Two paths to extreme seed indices then visit or skip every seed column. A skipped seed must descend toward an endpoint of the skipping link. Since its pair survived the low-tip test, that endpoint cannot be much lower than the seed, so the descent reaches it quickly. All selected seeds are consequently close to two short paths, where padding bounds their number.

Proposition 6.6 (Restricted packing bound). *Fix a narrow band J_2 . In any component of X_{J_2} , every closed d_{J_2} -ball of radius $4R$ meets seeds from at most*

$$M_{\text{pack}} = 2 + \lceil 1600R/g \rceil \tag{30}$$

groups. This bound is independent of the graph, the number of columns, and the history of the construction.

Proof. Let B be such a ball, with arbitrary center z . If it has no seeds there is nothing to prove. Otherwise fix $w_* \in B$ among the seeds. Only a narrow band contained in a broad band can have seeds. Its broad band J_1 is then unique, and the J_2 -paths from z to the seeds show that all of them lie in one J_1 -component.

All seeds of B have the same chart identifier in either system. Indeed $d(z, w) \leq 4R < 5$ for every such seed w , so z lies in the common chart of its eligible witness pair. Since charts partition X , those identifiers coincide. Write I for their common old system-A band and $s = d_I(o, \cdot)$ for its distance field. All center-to-seed paths of length at most $4R$, all descending witness paths, and the vertical intervals used below are contained in the relevant old chart band: they are at distance less

than 5 from an endpoint of an eligible witness pair. For example, a point on a center-to-seed path has distance at most $8R$ from w_* , and a descending path has length at most 0.02. The additional vertical intervals have length at most a_1 . In particular $J_1 \subset I$ as height intervals: the full column interval J_1 through any seed is within its radius-5 neighborhood.

Set $H = h(w_*)$ and form the profile $q_i = s(i, H)$, with $+\infty$ outside the old component as in Lemma 2.8. Put

$$t = s(w_*) - \frac{1}{2} 10^{-5}, \quad S = \{i : q_i \leq t\}. \quad (31)$$

For every seed $w \in B$, the path through z gives

$$|s(w) - s(w_*)| \leq 8R, \quad |s(w') - s(w)| \leq E_A.$$

The terminal column i of either chosen descending path for (w, w') therefore satisfies

$$q_i \leq s(w_*) + 8R + E_A - 10^{-5} + \epsilon + a_1 < t,$$

where we used $8R + E_A + \epsilon + a_1 < \frac{1}{2} 10^{-5}$. Thus S is nonempty, and it is connected in the active graph of I by Lemma 2.8.

Every column on any center-to-seed path of length at most $4R$ lies outside S . In fact, if such a path visits column i at point v , then

$$q_i \geq s(v) - a_1 \geq s(w_*) - 8R - a_1 > t.$$

All comparisons take place in I , as verified above. The noncrossing gap property, Lemma 2.5, now places all these path words and their endpoint representatives in one linear circular gap of S . Fix a representative of z to order this gap. Equivalent representatives also lie in the same gap, since a zero-length switch path satisfies the same avoidance argument.

Choose one representative seed from each group meeting B . In the linear order of the gap, choose a seed of minimum index among those to the left of the center index, if there is one, and similarly a seed of maximum index on the right. Let P_- and P_+ be shortest J_2 -paths from z to these extreme seeds. When one side has no seed, use the constant path at z on that side. Each path has length at most $4R$. For any chosen seed w , at least one of these two path words either visits its representative column or makes a switch strictly over its index. This is the elementary skipping property of Lemma 2.6, applied between the center and the extreme index on that side.

If the path visits the column of w , a vertical interval of length at most a_1 connects w to a point of that path in J_1 . Consider instead a switch strictly over its index. Let a, b be the switching columns, ordered within the gap so that the representative of w is in (a, b) . The switch is not a circular-neighbor link, because its strict interval within the gap contains a column and the complementary arc contains the nonempty set S . It is consequently a single switch at a tip e , by the column-model hypothesis. This tip is on P_- or P_+ , has height in J_2 , and satisfies

$$d(w, e) \leq 8R < 1. \quad (32)$$

The interval (a, b) is disjoint from S .

We claim that the witness partner w' has representative strictly outside the closed interval $[a, b]$ in the circle. If its representative were in (a, b) , its descending path, whose terminal column belongs to S , would have to visit column a or b . This follows from noncrossing, or directly from Lemma 2.4. Join that visit vertically to e , then follow the appropriate extreme path back to z and a shortest center-to- w path. The resulting path has length at most

$$0.02 + a_1 + 8R < 1,$$

contradicting $d(w, w') \geq 1$. If w' itself has representative a or b , the vertical connection to e gives the same contradiction without the descending segment. This proves the claim.

The link at e thus separates the two witness representatives strictly. Because the witness pair was not excluded and (32) holds, its failure to satisfy the barrier-drop condition implies

$$s(e) \geq s(w) - j. \quad (33)$$

The descending path from w must likewise hit column a or b , since it begins in (a, b) and ends in S . Let v be its first such hit and let p be the arclength to v . Lemma 6.3 and the vertical comparison with e give

$$s(v) \leq s(w) - p + \epsilon, \quad s(v) \geq s(e) - a_1 \geq s(w) - j - a_1.$$

It follows that $p \leq j + \epsilon + a_1$. The descending prefix and the vertical interval from v to e both lie in J_1 , so

$$d_{J_1}(w, e) \leq j + \epsilon + 2a_1. \quad (34)$$

We have proved that every chosen seed is within $\rho_{\text{pack}} = j + \epsilon + 2a_1$ in d_{J_1} of $P_- \cup P_+$. Distinct chosen groups have distinct width- g cells: their bands, band component, and two old identifiers have already been shown to coincide. Their seed points are radially padded, so Lemma 2.12 gives pairwise d_{J_1} -separation at least $b_0 = 0.01g$.

For each chosen seed, take one of the path points just constructed within ρ_{pack} , and assign it to one path containing it. Path points assigned to different seeds have d_{J_1} -distance at least $b_0 - 2\rho_{\text{pack}} \geq b_0/2$. Their arclength parameters on either path are therefore separated by at least $b_0/2$. A path of length $4R$ has at most $1 + 8R/b_0$ such points. The two paths together have at most

$$2 + 16R/b_0 = 2 + 1600R/g$$

assigned points. This proves (30). $\square$

Figure 3 shows the skipping-link argument and the short descending prefix that places a surviving seed near an extreme path.

Remark 6.7. The bounded number of nearby seed groups has two uses. Independent group coins can isolate one group's proposal at a pair, and only boundedly many families of random formulas can compete on a small ball. The first use gives secondary separation; the second preserves the common charts needed at later scales. The conclusion applies to seeds of the surviving witness pairs. It does not assert bounded overlap of all broad-band cells or their full neighborhoods: padded cells separate the chosen seeds, and exclusion of the component barriers brings those seeds close to the two short paths.

7 Ordered fields and separation on the two arcs

The packing bound limits the number of secondary proposals that can affect any one small ball. We now supply the scalar fields used by those proposals and by their final separation test. We use the circular order and the noncrossing links of the column model. All fields in this section are defined on a whole band X_I , including its separate components. A bound for vertical Lipschitz constants, together with compatibility at switches, implies the same bound for the intrinsic distance d_I wherever that distance is finite.

This section provides two tools. First, a point-distance function has a supporting field that is monotone toward a specified target along both circular arcs and retains the exact distance at that

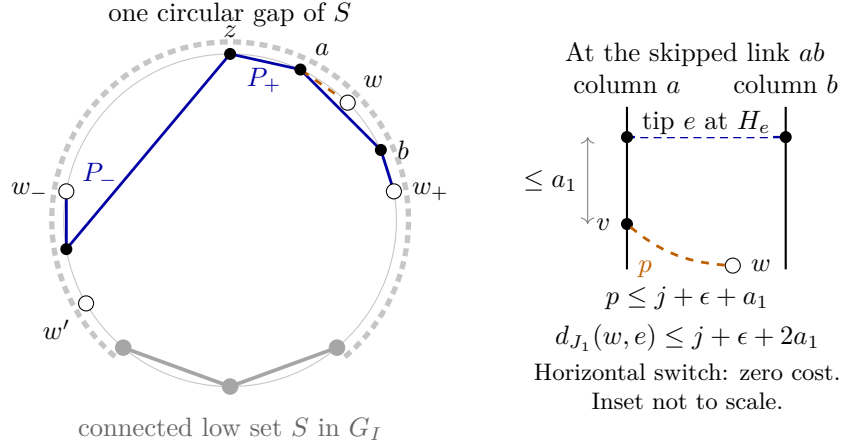


Figure 3: The key step in Proposition 6.6, shown for a surviving seed w . In a gap of the connected low set S , an extreme path skips the column of w by the terminal link ab . The witness partner w' is outside the small ball and lies strictly outside the closed interval $[a, b]$. The descending path from w must first reach a or b ; call that point v . Only this first prefix is shown in orange dashes. Survival of the low-tip test gives $s(e) \geq s(w) - j$, forcing the short prefix estimated in the inset. Blue paths use G_{J_2} ; connections of S may use the larger graph G_I . The circular drawing records column order, not metric lengths; extreme paths need not be monotone in that order.

target. Second, short paths on which the field rises and falls in a sufficiently thin band force a distant pair with nearly equal field values onto opposite arcs. A signed transverse field will then distinguish the pair.

7.1 Ordered supports of point distances

Definition 7.1. Let a, b be two distinct column indices. A real field s on X_I is *ordered from a to b* , with slope bound M , if it respects every switch, is M -Lipschitz on every vertical column, and, at each height $H \in I$, is nondecreasing on each of the two circular arcs oriented from a to b . In particular, the value on column a is a global minimum at that height, and the value on column b is a global maximum. We call the two open arcs the positive and negative arcs, with either assignment of these names.

Adding a constant to an ordered field preserves its order and slope bound. Multiplication by a positive constant multiplies the slope bound by that constant. Its negative is ordered after interchanging the two boundary columns. A field depending only on height is ordered for any choice of distinct boundary columns.

The supporting field below is obtained by taking running minima toward the target column on each arc. Connected distance sublevels are what make this operation compatible with the switches.

Lemma 7.2 (Ordered support of a point distance). *Suppose that k, w belong to one component of X_I . There is an ordered field L of slope bound 1 such that*

$$L(k) = 0, \quad L(w) = d_I(k, w), \quad L(v) \leq d_I(k, v) \quad (v \in X_I),$$

where the last inequality uses the convention that distance to a different component is infinite.

Proof. Put $D = d_I(k, w)$. If $D = 0$, the constant zero field suffices. If k, w have representatives on the same column, then $D = |h(w) - h(k)|$ and the field

$$L(v) = \min\{|h(v) - h(k)|, D\}$$

suffices, with any two distinct indices as boundary columns: vertical travel attains this distance on that column, and every path has length at least its absolute height change.

Otherwise choose a representative column a of k and a different representative column b of w . Write

$$f_i(H) = \min\{d_I(k, (i, H)), D\}, \quad H \in I,$$

where $f_i = D$ on any component not containing k . Every f_i is 1-Lipschitz, and the values respect switches. Moreover,

$$f_a(H) = \min\{|H - h(k)|, D\} = \min_i f_i(H). \quad (7.2.1)$$

For fixed H and $t \leq D$, every nonempty strict sublevel

$$S_t(H) = \{i : f_i(H) < t\}$$

is connected in the active column graph and contains a . To see this, let $i \in S_t(H)$ and take a path from k to (i, H) of length less than t . If the path visits column z at height T , its remaining length is at least $|T - H|$. Its prefix followed by vertical travel on z to height H therefore has length less than t . Thus every visited column belongs to $S_t(H)$, and projection of these paths to column indices connects $S_t(H)$ to a .

Consider a link with distinct endpoints p, q at height H , and put $c = f_p(H) = f_q(H)$. If neither endpoint is a , let $[p, q]_a$ be the closed circular interval from p to q that does not contain a . Then

$$f_z(H) \geq c \quad (z \in [p, q]_a). \quad (7.2.2)$$

Indeed, a strict-sublevel path from any column z with $f_z(H) < c$ to a avoids p, q , whereas Lemma 2.4 forbids such a path when z is strictly inside $[p, q]_a$. If one endpoint is a , both endpoint values equal the global minimum in (7.2.1).

Cut the circular order at a . For $i \neq a$, let $[i, b]_a$ denote the closed interval between i and b in the resulting linear order, and define

$$L(i, H) = \min_{z \in [i, b]_a} f_z(H), \quad L(a, H) = f_a(H). \quad (7.2.3)$$

These are finite minima of 1-Lipschitz functions. Along either arc from a to b the intervals in (7.2.3) shrink, so their minima are nondecreasing; the initial root value is a lower bound for all of them by (7.2.1).

We check switches explicitly. If a link meets a , its other endpoint has the global minimum value, and so does its value of L . Otherwise order its endpoints as $p < q$ in the order cut at a . By (7.2.2) the minimum of f on $[p, q]_a$ is their common value c . If b lies between p and q , both minima defining $L(p, H), L(q, H)$ equal c . If b lies outside that interval, the longer of the two minimizing intervals is the union of the shorter one and $[p, q]_a$. The minimum on the shorter interval is at most c , because it contains a link endpoint. The two minima are again equal. Thus L respects every switch and is an ordered field on the quotient.

Finally, $L(i, H) \leq f_i(H) \leq d_I(k, (i, H))$. At k , its root value is zero. At w , the minimizing interval is the singleton $\{b\}$ and its value is D . $\square$

7.2 A transverse coordinate

At a fixed height the ordered field increases from a to b along both arcs. Equal values can therefore occur on opposite arcs. The next field records the side and vanishes wherever a switch can change sides.

Definition 7.3. Let s be ordered from a to b with slope bound 4. In $\mathbb{R}^2$ define the set

$$Z_s = \{(s(a, H), H), (s(b, H), H) : H \in I\} \\ \cup \{(s(p, H), H) : H \in I \cap S_{pq}, p \text{ and } q \text{ lie on opposite open arcs}\}.$$

For a representative (i, H) define

$$t_s(i, H) = \begin{cases} \text{dist}_\infty((s(i, H), H), Z_s), & i \text{ on the positive arc,} \\ -\text{dist}_\infty((s(i, H), H), Z_s), & i \text{ on the negative arc,} \\ 0, & i \in \{a, b\}. \end{cases}$$

Here dist_∞ is distance for the maximum norm on $\mathbb{R}^2$.

Lemma 7.4. *The field t_s is well defined on X_I , respects all switches, and is 4-Lipschitz for d_I .*

Proof. The image of $H \mapsto (s(i, H), H)$ has Lipschitz constant at most 4 for the maximum norm, and distance to a fixed nonempty set is 1-Lipschitz. Thus t_s is vertically 4-Lipschitz on each open-arc column; it is identically zero on each boundary column. At a switch between columns on the same open arc, its values agree because both s and height agree. At a switch between opposite open arcs the common coordinate image belongs to Z_s , so both values vanish. At a switch involving a boundary column, its image belongs to Z_s for the same reason. These checks also prove independence of representatives. Summing along any finite path of vertical motions and switches proves the d_I bound. $\square$

7.3 Field excursions force opposite arcs

The next observation records the crossing argument with its height error. A path here can be a subpath of a concatenation; saying that it stays within length A of a point means that every point on it can be joined to that point by a path of length at most A in X_I . This condition holds, for example, for any subpath of the union of two paths of length at most A issuing from that point.

Lemma 7.5 (Overlapping traversals on one arc). *Let $J \subset I$ have width η . Suppose that s is ordered with slope bound 4, and that paths Γ_x, Γ_y in X_J use only columns of the same open arc. Assume that Γ_x stays within length A_x of x and Γ_y stays within length A_y of y . If the range of s on each path contains an interval $[u, v]$ with*

$$v - u > 32\eta,$$

then

$$d(x, y) \leq A_x + A_y + 2\eta.$$

Proof. Restrict and orient each path so that its endpoint values are u and v . Fix $H_* \in J$ and write $q_i = s(i, H_*)$. The values q_i are nondecreasing in the linear order on this open arc, and

$$|s(i, H) - q_i| \leq 4\eta \quad (H \in J). \tag{7.5.1}$$

Choose a point on Γ_x of value $c = (u + v)/2$, with representative column i . Then $|q_i - c| \leq 4\eta$. The initial column of Γ_y has reference value at most $u + 4\eta < q_i$, and its terminal column has reference value at least $v - 4\eta > q_i$. Its column word therefore starts strictly before i and ends strictly after i .

If Γ_y visits column i , connect the two visits on that column vertically at cost at most η , giving the claim. Otherwise Lemma 2.6 supplies a switch on Γ_y with endpoint columns $p < i < q$. If its height is $H_0 \in J$, then

$$|q_p - q_q| \leq 8\eta, \quad q_p \leq q_i \leq q_q.$$

In particular both endpoint reference values differ from q_i by at most 8η . The two endpoints of Γ_x have reference values respectively at most $u + 4\eta$ and at least $v - 4\eta$. Since $(v - u)/2 > 16\eta$, these lie strictly below q_p and strictly above q_q , respectively. Thus the middle point on column i is strictly inside the interval bounded by the link pq , whereas the endpoints of Γ_x are strictly outside it. Lemma 2.4 implies that Γ_x visits p or q . Connect this visit vertically to height H_0 and, if needed, use the zero-cost switch pq . This joins a point of Γ_x to a point of Γ_y at cost at most 2η (in fact η suffices when using the visited endpoint of that switch). The assumed connections to x, y complete the proof. $\square$

Lemma 7.6 (Quantitative separation on the two arcs). *Set $R = 10^{-8}$. Let s be an ordered field of slope bound 4 on X_I . Suppose that $x, y \in X_I$ satisfy*

$$1 \leq d(x, y) \leq 2, \quad d_I(x, y) = d(x, y), \quad |s(x) - s(y)| \leq 10^{-3}R.$$

Let $J \subset I$ be a common height interval of width

$$\eta \leq 10^{-6}R$$

containing x, y . For each $z \in \{x, y\}$, suppose that there are two paths γ_z^-, γ_z^+ in X_J , starting at z , of length at most $2R$, whose terminal points z^-, z^+ satisfy

$$s(z^-) < s(z) - 0.4R, \quad s(z^+) > s(z) + 0.4R.$$

Then $t_s(x), t_s(y)$ have opposite signs and

$$0.01R \leq |t_s(x)|, |t_s(y)| \leq 8.$$

Proof. Write $s_0 = s(x)$ and $H_x = h(x)$. We first prove that Z_s does not meet the rectangle

$$\mathcal{Q} = [s_0 - 0.2R, s_0 + 0.2R] \times [H_x - 0.02R, H_x + 0.02R]. \quad (7.6.1)$$

Suppose otherwise that $(c, H_0) \in Z_s \cap \mathcal{Q}$. At the reference height H_0 , the endpoint columns of each low path have field value strictly below c , and the endpoint columns of each high path have value strictly above c . Indeed their terminal values are respectively below $s_0 - 0.399R$ and above $s_0 + 0.399R$, and moving their values to height H_0 changes them by at most

$$4(0.02R + \eta) \leq 0.080004R.$$

These inequalities leave a strict margin from the interval $[s_0 - 0.2R, s_0 + 0.2R]$.

If (c, H_0) is the image of a boundary column, these strict inequalities contradict that column's extremality. Otherwise it is the image of a crosslink with one endpoint on each open arc. At height H_0 , monotonicity puts every column of value less than c on the root side of this crosslink and

every column of value greater than c on its opposite side. The projected concatenation $(\gamma_z^-)^{-1}\gamma_z^+$ must therefore visit an endpoint of the crosslink by Lemma 2.4. Each such visit is reachable from z in length at most $2R$, since it lies on one of the two original paths. Move vertically from the two visits to height H_0 and use the crosslink if needed. This produces an x - y path of length at most

$$4R + 2(0.02R + \eta) < 1,$$

contradicting $d(x, y) \geq 1$. This proves (7.6.1).

The maximum-norm ball of radius $0.01R$ around either $(s(x), h(x))$ or $(s(y), h(y))$ is contained in $\mathcal{Q}$. Hence

$$|t_s(x)|, |t_s(y)| \geq 0.01R. \quad (7.6.2)$$

In particular both centers have representatives on open arcs, with a well-defined choice of arc.

Consider the value strip

$$\mathcal{S} = \{v \in X_J : s_0 - 0.18R \leq s(v) \leq s_0 + 0.18R\}.$$

No representative of a point of $\mathcal{S}$ is on a boundary column, and no switch in $\mathcal{S}$ joins opposite open arcs. Either event would give an image in $Z_s \cap \mathcal{Q}$, because its height belongs to J and its value is within $0.18R$ of s_0 . Thus any path contained in $\mathcal{S}$ remains on a single open arc.

For $z = x, y$, the concatenation $(\gamma_z^-)^{-1}\gamma_z^+$ starts below the strip and ends above it. The scalar field is continuous along this path, including at switches. Take its first visit to the upper strip boundary, and its last visit to the lower strip boundary preceding that visit. The intervening subpath Γ_z stays in $\mathcal{S}$, traverses its full value interval, and lies on one open arc. Every point on it is reachable from z in length at most $2R$. If Γ_x, Γ_y were on the same arc, Lemma 7.5, with common interval of length $0.36R > 32\eta$, would give

$$d(x, y) \leq 4R + 2\eta < 1.$$

Consequently their two arcs are opposite.

It remains to relate these full traversals to the centers. From either center z , follow either one of its two original paths until its first exit from $\mathcal{S}$. The resulting initial subpath Π_z lies on the center's arc and its value range contains the interval between $s(z)$ and the strip boundary it first reaches. This interval has length at least $0.179R$. All its points are reachable from z in length at most $2R$. If the two centers had the same arc, exactly one of the opposite full traversals Γ_x, Γ_y would lie on that arc. If it were Γ_x , compare it with Π_y ; if it were Γ_y , compare it with Π_x . In either case the paths come from different centers and their value ranges overlap in an interval of length at least $0.179R > 32\eta$. Another application of Lemma 7.5 gives the same contradiction. Thus the centers have opposite arcs and, by (7.6.2), $t_s(x), t_s(y)$ have opposite signs.

Finally Lemma 7.4 and the assumed equality of band and unrestricted distances give

$$|t_s(x)| + |t_s(y)| = |t_s(x) - t_s(y)| \leq 4d_I(x, y) = 4d(x, y) \leq 8.$$

This proves both upper bounds. □

7.4 Ordered charts in system B

Besides ordinary charts, system B uses *ordered charts*. Such a chart has field

$$P = s + \lambda u,$$

where s is ordered with vertical Lipschitz constant at most 4 and u has Lipschitz constant at most K for d_I . At the creation of the chart, $u = 0$. The update in Section 8.2 changes u while preserving this bound. Our choice of λ gives $\text{Lip}(P) \leq 4 + \lambda K < 5$ for finite d_I , as required by the chart invariant of Section 3.

The band I , the ordered part s , its two boundary columns and choice of positive arc, and the label functions Z, σ remain fixed in original units for the lifetime of an ordered identifier. In particular its transverse field t_s remains fixed. Its region can shrink as other charts replace parts of it, and its auxiliary field u changes through the prescribed increments. Thus the current full field P may change without changing the identifier or the ordered data it records.

8 The secondary update

We now separate the pairs left by the two tests in Section 6. Their endpoints are seeds, grouped by their broad and narrow bands, their padded g -cell, and their two old chart identifiers. Proposition 6.6 bounds by M_{pack} the number of these groups meeting a narrow-band ball of radius $4R$. This lets us isolate one group's clipping at an endpoint with positive probability. In Section 9, the same bound will control the number of formulas competing on a small ball.

Clipping raises the field using a proposal based on distance to admitted seeds. In an ordinary chart it may also create an ordered chart, whose auxiliary field starts at zero. For a surviving pair in an ordered chart, either an endpoint is admitted in some orientation, allowing clipping, or each endpoint has short paths on which the fixed ordered part rises and falls. In the latter case Lemma 7.6 places the endpoints on opposite arcs, and the increment mode separates them using the transverse field. Clipping and increment are alternatives at the current scale; a newly created ordered chart is available for increments at subsequent scales.

8.1 Clipping near admitted seeds

Fix the orientation of the old B chart. In an ordinary chart its comparison field is $p_* = s$; in an ordered chart it is one of P and $-P$, chosen fairly and shared by the entire old identifier. For each group choose a representative w_c and set $p_c = p_*(w_c)$. The weak diameter bound of its g -cell, together with old-chart eligibility, gives

$$|p_*(w) - p_c| \leq 60g \quad \text{at every seed } w \text{ of the group.} \quad (35)$$

All short paths used for this comparison lie in the old field band.

In an ordinary chart choose an ordered support L with

$$L \leq p_*, \quad L(w_c) = p_c, \quad \text{Lip}_{\text{vert}}(L) \leq 1, \quad (36)$$

using Lemma 7.2 with the source of s and target w_c . For a constant chart use $L = 0$. All seeds in an ordinary chart are *admitted*. In an ordered chart admit w in the chosen orientation if

$$p_*(v) \geq p_c - 0.65R \quad \text{whenever } d_{J_2}(w, v) = R. \quad (37)$$

The test is vacuous when that sphere is empty. Admission supplies the boundary comparison needed by the shallow proposal below. Failure in both orientations will instead supply paths on which the comparison field rises and falls. These decisions use no fresh clipping noises.

A group's seeds have d_{J_1} -diameter at most $10g$, but may lie in several components of the narrow band. Give the group one independent fair coin, and keep its representative w_c , value p_c ,

orientation, and support L (when present) common to all those components. For a kept group, let F be its admitted seeds in one narrow-band component, provided this set is nonempty, and put

$$d_F(v) = \min_{w \in F} d_{J_2}(w, v).$$

Apply Theorem 4.1 to $0.9d_F$ on the entire component, with error E_B , obtaining Q_F . Use the resulting proposal only on $U_F = \{d_F \leq R\}$. Thus broad-band smallness controls the variation within a group, while narrow-band distance controls where its proposals can act. The shallow branch supplies the gain at seeds. In an ordinary chart the ordered-support branch protects a boundary where the old field decreases too fast. The height shield protects the band ends. Propose on this domain the minimum of

$$T_F(v) = p_c + 0.05R - Q_F(v), \quad (38)$$

$$O_F(v) = p_c + 0.05R + 4(L(v) - p_c) + \alpha_F, \quad \text{in an ordinary chart only,} \quad (39)$$

$$H_F(v) = \psi_F(h(v)). \quad (40)$$

Here α_F is uniform in $[0, E_B]$ and independent of the fresh interpolation noises defining Q_F . The height shield ψ_F is $p_c - 1 + \alpha_h$ at the two ends of J_2 , rises linearly to $p_c + 2 + \alpha_h$ within distance $0.09a_2$ of each end, and is constant on the intervening plateau; α_h is independently uniform in $[0, E_B]$. Extend it constantly beyond the band. Conditional on the old charts, bands, groups, orientations, admissions, and group coins, these interpolation and branch noises are fresh, and the noise families of different groups are independent. Replace p_* by its maximum with the proposals of all kept groups on their domains, and convert back to the old field orientation.

Lemma 8.1 (Validity of secondary clipping). *The secondary update preserves the nested-set and chart invariants. Its movement at each point, measured in symmetric difference, is bounded by a universal constant in scale units.*

Proof. First consider a point with $d_F(v) = R$. The shallow branch satisfies

$$T_F(v) \leq p_c + 0.05R - 0.9R + E_B = p_c - 0.849R.$$

In an ordered chart a nearest admitted seed gives $p_*(v) \geq p_c - 0.65R$, leaving a margin of $0.199R$. In an ordinary chart the same margin applies whenever that last inequality holds. Otherwise $L \leq p_*$ gives

$$O_F(v) - p_*(v) \leq 0.05R + 3(p_*(v) - p_c) + E_B < -1.899R.$$

Thus the proposal has a uniform margin below the old field on every distance boundary. At the height ends the shield gives a margin as well: each domain point lies within R of a seed, which lies within $10g$ in d of w_c , and the old comparison has Lipschitz constant less than 5 along these eligible paths. Its difference from p_c is less than $5(R + 10g) < 1/2$.

Every domain lies in its old chart, by seed eligibility. The comparison orientation and label variables are common on overlapping domains. Extend each proposal, after clamping against the old field, by the old field off its domain. The boundary margins and the clipping argument in Section 3 show that this extension glues vertically and respects switches. The finite maximum of these extensions has the same properties. The spatial branches have Lipschitz constant at most 4; the shield has constant $3/(0.09a_2)$, which is less than $1/\kappa_B$.

A winning atom of Q_F gives a constant plus or minus one point-source distance and hence a fresh ordinary chart on its narrow-band component. A winning O_F branch, after conversion back

to the old orientation, becomes the ordered part of a new chart on the old band, with $u = 0$. It has slope at most 4. Additive constants preserve its order; if conversion negates the comparison field, reverse the two boundary columns as in Definition 7.1. This ordered part, its band, and its arc assignment remain fixed under the new identifier; subsequent increments change only its auxiliary field. A height branch is rebased by Lemma 3.1. Unchanged points retain their chart, and uses of the same inherited atom in the same interpolation keep one identifier.

Finally Q_F lies within E_B of $0.9d_F$. Thus the shallow branch is at most $p_c + 0.051R$ and the old field is within $5(R + 10g)$ of p_c where the group can act. Taking a maximum over any number of groups does not add these bounds. Formula (5) gives a uniform symmetric-difference movement bound. $\square$

Lemma 8.2 (Separation at admitted seeds). *If a surviving pair has a seed admitted for one comparison orientation, then the no-change and secondary-clipping modes together give a probability bounded below by a positive universal constant of separating that pair by a positive universal amount.*

Proof. At an admitted seed w , $|Q_F(w)| \leq E_B$. Equations (35)–(36) imply that every applicable spatial branch exceeds $p_*(w) + 0.04R$: for the ordered branch use $|L(w) - p_c| \leq 10g$. The height shield is on its interior plateau. The other pair endpoint is outside this group's update domains, because the group's seeds have unrestricted diameter at most $10g$ and the pair distance is at least one.

At either endpoint all groups that could act have a seed within R in the narrow band. At most $2M_{\text{pack}}$ groups can therefore act in total at the two endpoints. Keeping the designated group and suppressing all other such groups has probability at least $2^{-(2M_{\text{pack}}+1)}$, a convenient lower bound. The desired ordered orientation, when needed, has probability $1/2$.

Let B_x, B_y denote the old system- B sets and B'_x, B'_y the updated sets. On this event one endpoint is unchanged, while the movement of the other has measure at least $0.04\kappa_B R$. If $\mu(B_x \triangle B_y) \geq 0.01\kappa_B R$, the no-change mode already gives the required separation. Otherwise the triangle inequality gives $\mu(B'_x \triangle B'_y) > 0.03\kappa_B R$, and in particular at least $0.02\kappa_B R$. Each mode has probability $1/4$, so both alternatives have uniform positive probability. $\square$

8.2 The increment in ordered charts

For an ordered identifier, the ordered part s , its band I , and its arc assignment stay fixed while the auxiliary field u changes and the current region $\mathcal{R}$ may shrink. Its transverse field t_s is therefore fixed as well. We define the increments in original length units, so these stored fields do not change when a step is analyzed in units of its scale.

Define a continuous trapezoidal function $H : [0, \infty) \rightarrow [0, 1]$ which vanishes off $[0.005R, 10]$, equals one on $[0.01R, 8]$, and is linear on the two remaining intervals. At scale r , in original units, add to u the function

$$\pm t_s(v)H(|t_s(v)|/r) \min\{1, d(v, X \setminus \mathcal{R})/r\}. \quad (41)$$

The sign is fair and common to the old identifier. When the complement is empty its distance factor is one. The formula is defined on the chart's whole band, with changes to the sets effective only in its current region. Old terms of u are retained without modification.

Lemma 8.3 (Increment bound). *If $Q > 2000/R$, all increments of one ordered identifier together have d_I -Lipschitz constant at most $K = 100$. They preserve the nested sets, introduce no chart interfaces, and move each point by at most a universal constant times the scale.*

Proof. The function t_s is 4-Lipschitz in d_I . The scalar function $t \mapsto tH(|t|/r)$ has Lipschitz constant at most 6 and magnitude at most $10r$. The distance factor has Lipschitz constant at most $1/r$ for d , hence for d_I . Each increment has Lipschitz constant at most $24 + 10 = 34$.

Its support is contained in $0.005Rr < |t_s| < 10r$. Since t_s is fixed for this identifier, the support intervals at distinct scheduled scales are disjoint when $Q > 2000/R$. In a comparison of two points at most two summands are nonzero at either endpoint. Subtracting their zero values at the other endpoint if necessary bounds the difference by $68d_I(v, w)$. Thus $\text{Lip}(u) \leq 68 < K$ at all times.

The distance factor makes the movement vanish continuously at the region boundary. Inside the region the new field has Lipschitz constant at most $4 + \lambda K < 5 < 1/\kappa_B$. The vertical gluing argument applies, and the region partition does not change. The amplitude bound is $10r$ before the factor λ , proving the last assertion. $\square$

Lemma 8.4 (The remaining pairs). *Suppose a surviving pair has no admitted endpoint in either ordered orientation. Then the increment mode separates it by a positive universal amount with uniformly positive probability.*

Proof. Normalize the current scale to one. Ordinary charts admit every seed, so the old B chart is ordered. Failure of each admission test supplies a point on the corresponding d_{J_2} -sphere of radius R and an attaining path of length R . Failure for P supplies a decrease of P , and failure for $-P$ supplies an increase. By (35), each change has magnitude greater than $0.65R - 60g$. The correction λu changes by at most λKR on either path. Thus the ordered part s has a positive and a negative change of magnitude greater than

$$0.65R - 60g - \lambda KR > 0.4R$$

along these paths from each endpoint.

A shortest x - y path has length at most 2 and stays in the eligible radius-5 ball, hence in the old band I . This proves $d_I(x, y) = d(x, y)$ and bounds the change of u by $2K$. The B cutoff therefore gives

$$|s(x) - s(y)| \leq 10^{-4}R + 2\lambda K < 0.001R.$$

The four excursion paths lie in the common band J_2 , whose width satisfies Lemma 7.6. Applying that lemma gives opposite signs of t_s and $0.01R \leq |t_s(x)|, |t_s(y)| \leq 8$.

These bounds put both endpoints on the plateau of the trapezoidal factor. Eligibility also puts their radius-5 balls in the current region, so the distance factor in (41) equals one at both endpoints. For one of the two increment signs,

$$|P_{\text{new}}(x) - P_{\text{new}}(y)| \geq \lambda |t_s(x) - t_s(y)| \geq 0.02\lambda R.$$

Formula (5) proves separation at least $0.02\kappa_B\lambda R$. The favorable sign has probability at least $1/2$. $\square$

9 Preserving charts and assembling the embedding

The local constructions now provide a uniform chance of separation at a pair's distance scale. Two estimates let us use that chance in a complete schedule: coarser scales rarely destroy the common charts required by the test, and finer scales move too little label mass to erase its certificate. We first collect these estimates, then choose the scale spacing and write the resulting finite cut embedding.

9.1 The one-step estimates

All constants in this section depend only on the fixed choices in Section 5.1. In particular they are independent of the number of columns, band components, chart identifiers, and scales.

Proposition 9.1 (Movement and success). *There are constants $M < \infty$ and $b, p > 0$ such that the following hold at every step of scale r .*

- (i) *At every point of X , in either system, the symmetric-difference movement of the unrefined set is at most Mr .*
- (ii) *Given any past for which a vertex pair at distance in $[r, 2r)$ is eligible, the step has probability at least p of producing one of these certificates:*
 - (a) *unrefined symmetric-difference separation at least br in one system;*
 - (b) *a non-neighbor tip e and a set of labels of measure at least br in system A on which both pair vertices are active and e is inactive, with the vertices strictly on opposite sides of that link.*

Proof. Lemma 6.1 bounds movement in the main clipping. Secondary movement is bounded by Lemma 8.1; increments are bounded by Lemma 8.3. These estimates apply even at points outside the finite vertex list. There are no simultaneous main updates at a point except at cell overlaps, where every update is unchanged. Secondary updates take a maximum, whose movement is bounded without summing over groups. Taking the largest of the bounds gives M .

Work in scale units. If one of the cutoffs in (19) fails, no change preserves separation at least the minimum of $10^{-4}a_2, \kappa_A E_A, \kappa_B 10^{-4}R$. Otherwise the interior requirements hold with probability at least $1/4$. Conditional on the bands and cells, a missing descending path is handled by Corollary 6.4: keep that endpoint's cell and suppress the other, giving positive separation with probability at least $1/4$ conditional on the main mode. If both paths exist, an excluded pair has the component certificate on a label set of measure at least $\kappa_{Aj}/4$, which the no-change mode preserves.

For a surviving pair, admission at an endpoint is handled by Lemma 8.2, including its no-change alternative. If neither orientation admits either endpoint, apply Lemma 8.4. Each case uses one of four modes, at most one fair orientation, and at most $2M_{\text{pack}} + 1$ prescribed group coins. It follows, for example, that $p = 2^{-(2M_{\text{pack}}+10)}$ is a permissible lower bound after all interior and mode probabilities. Choose $b > 0$ below all the finitely many displayed gains and certificate masses. These choices depend only on the fixed parameters. $\square$

9.2 The probability of splitting a small ball

The following estimate concerns chart identifiers, not cut components. System A is refined only after all the updates.

Proposition 9.2 (Chart persistence). *There exist constants $C_0 < \infty$ and $\rho_0 > 0$ such that the following holds. Condition on the past at a step of scale r . Suppose a fixed closed ball $B_d(v, \rho r)$ lies in a single old region in each system. For $0 < \rho \leq \rho_0$, the probability that this ball is split between different new regions in either system is at most*

$$C_0(\rho + \eta_0). \tag{42}$$

Here η_0 is the relative mesh allowance from Section 5.2.

Proof. Normalize $r = 1$ and write $B = B_d(v, \rho)$. Every point of B has a shortest path to v contained in B . Joining two such paths gives a path of length at most 2ρ between any two points of B . We will use these paths for all band-distance comparisons.

There are two ways an update could give different chart identifiers on B : its domain could end inside the ball, or different formulas could win there. Strict margins at domain boundaries prevent the first kind of change. The comparison estimate of Corollary 4.6 controls the second.

Height bands. The probability that an endpoint of either random height tiling lies within ρ of $h(v)$ is $O(\rho + \eta_0)$. Exclude this event and condition on the tilings. The ball lies wholly in one component of a broad band and one component of a narrow band: all its center paths are paths in those bands. For clipping, each such band is either processed in its entirety or left unchanged. A narrow band not contained in a broad band has no secondary seeds, so secondary clipping also leaves it unchanged.

Main clipping. Condition further on the radial cells and their coins. Suppose first that B meets more than one width- D cell, including through different representatives of a quotient point. For any touched cell C and any $w \in B \cap C$, a path through the center to a point in another cell reaches a frontier point $z \in B \cap \partial C$ within length 2ρ . If C is available, its old comparison field is the common old field on B . Thus

$$F_C(w) \leq s(z) + (1 - \zeta_A)d_{J_1}(w, z) - 3E_A \leq s(w) + 4\rho - 3E_A.$$

For $\rho_0 < E_A/10$ the proposal cannot raise the field anywhere on $B \cap C$. Unavailable cells are unchanged as well. Hence the whole ball retains its old identifier. In particular we do not need to exclude a random radial boundary crossing.

It remains to consider a ball lying in one cell C . If that cell is unavailable or its coin is suppressed, it is unchanged. Otherwise its comparison expression on B is

$$\max\{s, \min\{F_C, p_{h,C}\}\}.$$

Here $F_C = \tilde{p}_C - 4E_A$ is one interpolation construction, $p_{h,C}$ is a fixed height function plus its independent offset, and s is the only branch with no fresh noise. The input p_C , portals, and bands have all been fixed before these noises are sampled. Every center path has length at most ρ in the interpolation band and in the old chart band. Corollary 4.6, with the fixed Lipschitz bound of the height function, therefore gives one winning branch and, if that branch is the interpolation, one named atom throughout B , except with conditional probability $O(\rho + \eta_0)$. A common old branch retains its identifier; a common spatial atom or height branch gives a common new chart. At interpolation layer seams the inherited atom keeps its name, as required by the interpolation theorem.

Secondary clipping. Condition on the cells, surviving groups, comparison orientations, admissions, and group coins. These choices precede the fresh clipping noises. A group whose domain meets B has an admitted seed at band distance at most $R + 2\rho < 4R$ from v . Proposition 6.6 therefore bounds the number of such groups by M_{pack} .

If a group's domain meets but does not contain B , its boundary is reached within length 2ρ from every point of its domain in the ball. Lemma 8.1 puts the proposal below the old field at this spatial boundary by a fixed positive margin; height-boundary crossings have already been excluded. The fixed Lipschitz constants of the proposal branches and the old field extend this

nonaction to the domain's intersection with the ball when ρ_0 is small enough. Outside the domain there is no update by definition. Discard such a group. Every remaining relevant domain contains B , so its proposal is described by one fixed expression throughout the ball.

Each relevant domain lies in its seed's old chart and meets B . Since regions partition X , that chart must be the ball's common old chart. Thus all these proposals use the same old comparison orientation and label variables. The ball lies in one narrow-band component, so each group contributes only the interpolation for that component. Their combined expression is the maximum of p_* and the group proposals, each proposal being the minimum of T_F , H_F , and, in an ordinary chart, O_F . The T_F are signed, shifted interpolation functions; the H_F and O_F have independent additive offsets. The only input without fresh noise is the old field p_* . The group data and the interpolation inputs are already fixed, and fresh noise families are independent between groups. All center paths are permitted in their narrow-band metrics and in the old chart band.

Apply Corollary 4.6 to this expression. Each interpolation contributes its bounded list of candidate atom names; the number of lists is bounded by M_{pack} . All constants are therefore uniform in the graph and the past. Except with conditional probability $O(\rho + \eta_0)$, one branch and, where applicable, one atom name wins throughout B . The resulting chart identifier is common to the ball. Repeated appearances of an inherited atom use its interpolation and original atom name, so crossing a portal collar does not create another identifier.

No-change and increment modes introduce no chart identifiers. Taking ρ_0 smaller than the finitely many bounds used above and summing the exceptional probabilities proves (42). $\square$

9.3 Choice of spacing and the final map

Choose $Q = 2^m$ large enough for the support condition in Lemma 8.3, for $20/Q < \rho_0$, and for

$$\frac{40C_0}{Q-1} < \frac{1}{2}, \quad \frac{6M}{Q-1} < \frac{b}{2}. \quad (43)$$

Choose the mesh allowance so that $\eta_0 \leq r_f/r$ for every pair of scales $r_f < r$ in the finite collection of schedules. Then the mesh term in (42) is no larger than the corresponding scale ratio. This choice does not change any of the constants above.

Fix a pair and its designated scale r_f , and consider $B_d(x, 20r_f)$. Initially it has one chart in each system. Conditional on having retained that property up to a coarser scale r , its first loss probability is at most $40C_0r_f/r$. Therefore

$$\Pr[\text{a first loss before } r_f] \leq 40C_0 \sum_{k \geq 1} Q^{-k} < \frac{1}{2}.$$

On the complementary event the pair is eligible at scale r_f . Furthermore any D -cell needed for its main update has its entire 1-neighborhood inside this ball: its weak diameter is at most $10D$. The old chart bands contain the short paths used in all tests.

By Proposition 9.1, conditional on this event the scale produces a certificate with probability at least p . Subsequent unrefined movement at any one point is deterministically at most

$$M \sum_{k \geq 1} r_f Q^{-k} = \frac{Mr_f}{Q-1}.$$

An ordinary pair separation loses at most twice this quantity. For a component certificate, remove labels whose membership later changes at either vertex or at its tip; their total measure is at most

three times this quantity. At least $br_f/2$ of the original certificate therefore remains. Lemma 3.2 gives final separation for all those labels, regardless of changes elsewhere. In either case the final sum of the refined A distance and the unrefined B distance is at least $br_f/2$ on the successful event. Its expectation in the designated schedule is consequently at least

$$\frac{1}{2}p \frac{br_f}{2} \geq \frac{pb}{8}d(x, y).$$

The designated schedule is the residue class containing the dyadic exponent of r_f ; every other schedule contributes a nonnegative distance. Mix uniformly over the m schedules. Within each schedule use the probability law of the random construction, weighting each outcome by its probability. More precisely, an outcome is a branch of the finite execution tree, and its weight is the product of the conditional probabilities of the choices along that branch. These weights sum to one even though later geometric choices depend on earlier outcomes. Take the finite direct sum of their measure spaces with these weights, and of the two systems. Subtract the assignment at a fixed reference vertex in every summand. By Lemma 3.2 the resulting map F is into a real L_1 space and satisfies

$$\frac{pb}{8m}d(x, y) \leq \|F(x) - F(y)\|_1 \leq 2d(x, y) \quad (x, y \in V).$$

This map can be written using finitely many cut coordinates. For every final outcome and every label in system B , record the subset of V that is active. For system A , record separately, for each active column component, the subset of original vertices active in that component. Each such subset S contributes the cut distance $|\mathbf{1}_S(x) - \mathbf{1}_S(y)|$. Integrating over the labels and averaging with the schedule and outcome weights therefore gives numbers $w_S \geq 0$ such that

$$\|F(x) - F(y)\|_1 = \sum_{\emptyset \neq S \subseteq V} w_S |\mathbf{1}_S(x) - \mathbf{1}_S(y)|.$$

Empty and full subsets contribute zero and have been discarded. Every remaining weight is finite: if $x \in S$ and $y \notin S$, its contribution is bounded by the already established upper bound $2d(x, y)$. Hence

$$v \mapsto (w_S \mathbf{1}_S(v))_{\emptyset \neq S \subseteq V}$$

is a map into a finite-dimensional real ℓ_1 space with exactly the same distances as F .

The column realization identifies these distances with the original d_G . Rescale F by $8m/(pb)$ to obtain Theorem 1.1 with $C = 16m/(pb)$. If V has a single point use the zero map. Every constant is independent of the graph size and of its positive edge lengths.

10 A consequence for almost-embeddable families

Lee and Sidiropoulos reduce their fixed-parameter almost-embeddable families to planar and bounded-treewidth pieces joined at single vertices [13, Theorem 1.7]. We combine this reduction with Theorem 1.1 and the bounded-treewidth embedding theorem of the companion [17, Theorem 1.1].

A 1-sum joins two graphs by identifying one vertex of each. A stochastic embedding of distortion D is a distribution of maps into graph metrics that do not contract distances, with expected expansion at most D for every pair.

Informally, the construction of these families starts with a graph drawn on a compact surface of genus g , orientable or nonorientable. One adds k attachments of prescribed width w , called

fringes, to k faces, and then a additional vertices with arbitrary adjacencies; these last vertices are the apices. In particular, when $k = 0$ this includes bounded-genus graphs with a bounded number of arbitrarily attached vertices. The corollary uses Lee and Sidiropoulos’s named family and their prescribed attachment and width conventions.

Corollary 10.1 (Fixed almost-embeddable families). *Fix nonnegative integers g, k, w, a , and interpret $\text{AE}(g, (k, w), a)$ exactly as the almost-embeddable family of Lee and Sidiropoulos [13, Theorem 1.7]. Let $h_0 = h_0(g, k, w, a) \in \mathbb{N}_0$ be the target treewidth bound and let $D = D(g, k, w, a) \geq 1$ be a finite stochastic distortion bound supplied there. Put*

$$\hat{h} = \max\{h_0, 1\}, \quad K(g, k, w, a) = D \max\{C_{\text{pl}}, C_{\hat{h}+1}\},$$

where C_{pl} is the constant of Theorem 1.1, and C_t is the companion’s constant for bags of cardinality at most $t \geq 2$. Every finite connected simple graph G in this named family, with strictly positive real edge lengths, admits a map f into a real L_1 space satisfying

$$d_G(u, v) \leq \|f(u) - f(v)\|_1 \leq K(g, k, w, a) d_G(u, v) \quad (u, v \in V(G)).$$

For the same underlying graph, $K(g, k, w, a)$ also bounds the undirected fractional concurrent flow–cut gap ϕ_*/λ_* for the finite nonnegative capacities and demands defined in Section 1.1, whenever $\lambda_* > 0$.

Proof. The singleton case is immediate. Otherwise, replace each input edge length by the distance between its endpoints; this preserves the metric and gives reduced positive lengths. Lee–Sidiropoulos Theorem 1.7 stochastically embeds this metric with distortion at most D into finite 1-sums of planar graphs and graphs of treewidth at most h_0 . Since the companion indexes C_t by bag cardinality, enlarging h_0 to $\hat{h}$ gives the applicable constant $C_{\hat{h}+1}$. Put $M = \max\{C_{\text{pl}}, C_{\hat{h}+1}\}$.

For a connected target piece with nonnegative edge lengths, add $\varepsilon > 0$ to every edge and apply Theorem 1.1 or the companion theorem. As $\varepsilon \downarrow 0$, the shortest-path matrices converge, so the corresponding L_1 distance matrices have a subsequential limit in the closed finite cut cone, retaining exactly the bound M and identifying zero-distance pairs. The finite expected expansion ensures that all images of the finite input lie in a single finite-distance target component almost surely. Lemma 1.5 of [13] retains the bound M on finite 1-sums, and its Lemma 1.2 composes this bound with the stochastic distortion D , giving $K = DM$. Applying the same cut-cone limit to input lengths, followed by the fixed-graph equality of [7, Section 3, Theorem 3.2], gives the flow–cut conclusion. $\square$

This corollary concerns only the cited fixed-parameter almost-embeddable families. It does not establish closure under general clique sums of order greater than one, and therefore does not prove the full GNRS conjecture.

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